House Rent Prediction Dataset

MAGICBRICKS

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Introduction:

This econometrics project explores the dynamics of house rental prices using real-world data sourced from the Kaggle dataset titled *"House Rent Prediction."* The primary goal is to apply econometric techniques to analyze and predict house rent values based on various influential factors such as location, size, furnishing status, and number of rooms. The structure of this project follows a systematic approach aligned with the course requirements.

The report is organized into six main sections. It begins with an **Introduction**, which presents the background of the study and outlines the research questions. The **Methodology and Analysis** section describes the dataset, details the data cleaning procedures, and presents the econometric models used to investigate the relationships among variables. It also includes assumption testing and interpretation of results. The **Conclusion** offers a brief summary of findings in simple terms suitable for a general audience. **References** used throughout the report are listed in a dedicated section, and the **Appendices** include all relevant Python codes and supplementary outputs that support the analysis.

Through this structured investigation, the project aims to provide insights into the determinants of rental prices and the predictive power of econometric modeling in housing markets.

Research Objective:

The phenomenon under investigation is the variation in **residential rental prices** across different Indian cities and property characteristics. House rent is a critical economic and social issue, especially in densely populated and rapidly urbanizing countries like India. It not only affects household budgeting and living standards but also reflects broader trends in urban development, housing demand, and affordability. Understanding the determinants of rental prices helps policymakers, real estate developers, and tenants make informed decisions. With rising urban migration and housing shortages in many cities, analyzing what drives rent levels and how these factors vary spatially is more relevant than ever.

The main objective of this project is to identify and quantify the key factors that influence house rent prices. Specifically, the study aims to address three core research questions:

1. **What are the key factors influencing house rent prices, and which factors have the most significant impact?**
2. **Does the size of the property affect the rent price differently across different cities?**
3. **Are there differential effects of area type (Carpet vs. Super Area) on rent prices across cities?**

By answering these questions through econometric modeling, this study seeks to uncover patterns and insights that can contribute to a better understanding of urban housing economics in India.

Methodology and Analysis:

Dataset Description:

The dataset used in this project is sourced from Kaggle and titled *"House Rent Prediction Dataset"*, consists of **4,746 rental property** listings across several major Indian cities **in 12 columns**. It includes various quantitative and categorical features related to property characteristics, location, and listing details. These variables provide a comprehensive foundation for conducting econometric analysis to explore the factors influencing rental prices. The data was retrieved from MagicBricks, a specialized online platform for buying, selling, and renting residential properties in India.

In this study, the **dependent variable** is:

- **Rent**: A continuous variable representing the monthly rental price of the property in Indian Rupees (INR).

The **independent variables** include a mix of continuous, discrete, and categorical variables:

1. **Posted On:** (Date as string format) the date on which the rental listing was posted. This variable is **not used** as a predictor in this study, but it indicates the temporal aspect of the listing.
2. **BHK:** A discrete (integer) variable showing the number of bedrooms, halls, and kitchens in the property.
3. **Size:** A continuous variable indicating the total area of the property in square feet.
4. **Floor**: Categorical variable initially presented as a text string (e.g., "2 out of 10"), this variable will be processed into numerical format to extract the actual floor level **(floor)** and total number of floors **(no_of_floors)** in the cleaning process.
5. **Area Type:** A categorical variable describes how the area is measured. Takes values of Super Area or Carpet Area and Build Area.
6. **Area Locality:** Text refers to the specific neighborhood or sub-area within a city where a property is located. It's typically helps identify the exact region or zone within an urban area. This variable was **execluded** from our study.
7. **City:** A categorical variable identifying the city in which the property is located.
8. **Furnishing Status:** A categorical variable describing whether the house is furnished, semi-furnished, or unfurnished.
9. **Tenant Preferred:** A categorical variable that indicates the type of tenants preferred by the property owner (e.g., families, bachelors, or both).
10. **Bathroom:** A discrete (integer) variable representing the number of bathrooms in the propery.
11. **Point of Contact:** A categorical variable that shows whom should you contact for more information regarding the Houses/Apartments/Flats.—Contact Agent or Contact Owner and Contact Builder.

This dataset is particularly suitable for econometric analysis due to its **richness, multidimensionality, and real-world relevance**. It contains a balanced mix of **quantitative** and **categorical** variables that reflect essential dimensions of the rental housing market in India. As described, the dataset includes **structural attributes** such as **Size**, **BHK**, and **Bathroom**, which directly represent the physical characteristics of properties. It also captures **locational aspects** through the **City** variable, and originally included **Area Locality**, which—although excluded from modeling—demonstrates the granularity and spatial coverage of the data.

Moreover, the dataset incorporates **market classifications** such as **Furnishing Status**, **Area Type**, **Tenant Preference**, and **Point of Contact**, which represent qualitative differences in how rental listings are offered and targeted. These variables provide valuable context for understanding pricing behavior and tenant segmentation. For example, differences in rent between **Furnished** and **Unfurnished** properties, or between **Carpet Area** and **Super Area**, are central to the kinds of comparative questions posed in this study.

Thus, the dataset's **diverse variable types and detailed structure** make it highly suitable for testing hypotheses on **rental price variation**, **spatial pricing efficiency**, **regional disparities**, and **measurement-based valuation differences**. It supports both main-effect and interaction-based econometric models that align directly with the study's objectives.

Descriptive Statistics and Data Visualization:

Before proceeding to model estimation, we performed an extensive exploratory data analysis to understand the characteristics and distribution of key variables. First, we conducted a structural inspection of the dataset and removed irrelevant or redundant entries, such as listings with missing or malformed floor information, properties classified under "Built Area," or listings that used "Contact Builder" as a point of contact. Outliers were identified and excluded using Cook's distance to ensure robustness and reliability in the subsequent regression models.

To visually assess the relationships between numerical variables and rent, several scatterplots were generated. These included plots of rent against property size, number of bedrooms (BHK), number of bathrooms, floor level, and total number of floors. Each scatterplot provided preliminary insights into linear and nonlinear patterns, particularly highlighting that rent tends to increase with size and amenities like additional bathrooms or higher floors.

Boxplots were also constructed for categorical variables such as **City**, **Area Type**, **Furnishing Status**, **Tenant Preferred**, and **Point of Contact**, with the logarithm of rent as the dependent variable. These visualizations revealed substantial variation in rental prices across different cities and furnishing statuses, as well as suggestive differences between area types like Carpet Area and Super Area. The log transformation of rent was applied to stabilize variance and better meet normality assumptions in later regression analysis.

Overall, the descriptive analysis highlighted key trends and variability in the data, provided intuition for variable selection, and supported the formulation of hypotheses to be tested in the econometric models.

Data Cleaning:

```
> str(data)
'data.frame':	4746 obs. of  10 variables:
 $ BHK              : int  2 2 2 2 2 2 2 1 2 2 ...
 $ Rent             : int  10000 20000 17000 10000 7500 7000 10000 5000 26000 10000 ...
 $ Size             : int  1100 800 1000 800 850 600 700 250 800 1000 ...
 $ Floor            : chr  "Ground out of 2" "1 out of 3" "1 out of 3" "1 out of 2" ...
 $ Area.Type        : chr  "Super Area" "Super Area" "Super Area" "Super Area" ...
 $ City             : chr  "Kolkata" "Kolkata" "Kolkata" "Kolkata" ...
 $ Furnishing.Status: chr  "Unfurnished" "Semi-Furnished" "Semi-Furnished" "Unfurnished" 
...
 $ Tenant.Preferred : chr  "Bachelors/Family" "Bachelors/Family" "Bachelors/Family" "Bach
elors/Family" ...
 $ Bathroom         : int  2 1 1 1 1 2 2 1 2 2 ...
 $ Point.of.Contact : chr  "Contact Owner" "Contact Owner" "Contact Owner" "Contact Owner
```

Upon examining the dataset, it is evident that several variables should be converted to categorical types, as they represent distinct categories rather than nominal values. These include Area.Type, City, Furnishing.Status, Tenant.Preferred, and Point.of.Contact. Converting them to categorical format will enhance model performance and optimize memory usage. Additionally, the Floor variable contains compound information (e.g., "Ground out of 2", "1 out of 3") and should be split into two separate columns: one for the floor number and another for the total number of floors in the building. This transformation will facilitate better analysis.

```
> summary(df)
      BHK             Rent              Size          no_of_floors        floor
 Min.   :1.000   Min.   :   1200   Min.   :  10.0   Min.   : 1.000   Min.   :-2.000
 1st Qu.:2.000   1st Qu.:  10000   1st Qu.: 550.0   1st Qu.: 2.000   1st Qu.: 1.000
 Median :2.000   Median :  16000   Median : 850.0   Median : 4.000   Median : 2.000
 Mean   :2.084   Mean   :  34993   Mean   : 967.5   Mean   : 6.973   Mean   : 3.436
 3rd Qu.:3.000   3rd Qu.:  33000   3rd Qu.:1200.0   3rd Qu.: 6.000   3rd Qu.: 3.000
 Max.   :6.000   Max.   :3500000   Max.   :8000.0   Max.   :89.000   Max.   :76.000
                                                    NA's   :4
        Area.Type           City          Furnishing.Status          Tenant.Preferred      Bathroom
 Built Area :   2   Bangalore:886   Furnished     : 680    Bachelors       : 830    Min.   : 1.000
 Carpet Area:2298   Chennai  :891   Semi-Furnished:2251    Bachelors/Family:3444    1st Qu.: 1.000
 Super Area :2446   Delhi    :605   Unfurnished   :1815    Family          : 472    Median : 2.000
                    Hyderabad:868                                                   Mean   : 1.966
                    Kolkata  :524                                                   3rd Qu.: 2.000
                    Mumbai   :972                                                   Max.   :10.000

        Point.of.Contact
 Contact Agent  :1529
 Contact Builder:   1
 Contact Owner  :3216
```

The categorical variables (Area.Type, City, Furnishing.Status, Tenant.Preferred, and Point.of.Contact) were successfully converted to factor type. Additionally, the original Floor column was split into two separate numeric variables: floor and no_of_floors. During this process, it was observed that there are **4 missing values** in the no_of_floors variable, as seen in entries like the one shown below (see attached image). These missing values were removed to avoid complications in downstream modeling. Furthermore, both floor and no_of_floors are now treated as numeric variables, with special encoding applied: -1 for **Upper Basement**, -2 for **Lower Basement**, and 0 for the **Ground Floor**.

4492	3	15000	900	1	Super Area	Hyderabad	Semi-Furnished	Bachelors/Family	3	Contact Owner

Moreover, sparse categories such as *"Contact Builder"* in Point.of.Contact (only 1 observation) and *"Built Area"* in Area.Type (only 2 observations) were excluded. These sparse categories would result in zero-only columns in the design matrix (X), potentially causing matrix singularity issues and preventing the inversion process, which is crucial for various linear estimation methods.

Furthermore, there are clear outliers present in several numerical variables, specifically in Rent, Size, and Bathroom. One particularly inconsistent observation lists a flat with 10 bathrooms, while the number of bedrooms (BHK) is only 1—a configuration that is highly unrealistic in the context of residential apartments and strongly suggests a data entry error. Based on listings from MagicBricks, it is likely that the correct number of bathrooms was 1, not 10. Therefore, all observations with 6 or more bathrooms were removed, as these are likely to be typographical errors rather than legitimate listings.

BHK	Rent	Size	Floor	Area Type	City	Furnishing Status	Tenant Preferred	Bathroom	Point of Contact
1	200000	8000	Ground out of 4	Super Area	Hyderabad	Unfurnished	Bachelors/Family	10	Contact Owner

To further address the presence of outliers, **Cook's Distance** was applied—a widely used diagnostic method to identify **influential observations**, The Cook's Distance for the *i*-th observation is computed as:

$$C_i = \frac{\sum(\hat{y_j} - \widehat{y_{j(i)}})^2}{\widehat{\sigma^2}(k+1)}, i = 1,2,3, \dots, n$$

Observations with a Cook's Distance less than the threshold of **4/n** (where *n* is the total number of observations) were considered non-influential outliers and thus removed from the dataset.

```
> summary(df_clean)
      BHK             Rent              Size          no_of_floors        floor       
 Min.   :1.000   Min.   :  3000   Min.   :  10.0   Min.   : 1.000   Min.   :-2.000  
 1st Qu.:2.000   1st Qu.: 10000   1st Qu.: 550.0   1st Qu.: 2.000   1st Qu.: 1.000  
 Median :2.000   Median : 15000   Median : 840.0   Median : 4.000   Median : 2.000  
 Mean   :2.046   Mean   : 29900   Mean   : 932.2   Mean   : 6.636   Mean   : 3.307  
 3rd Qu.:3.000   3rd Qu.: 30000   3rd Qu.:1200.0   3rd Qu.: 6.000   3rd Qu.: 3.000  
 Max.   :6.000   Max.   :650000   Max.   :4500.0   Max.   :89.000   Max.   :76.000  
         Area.Type           City          Furnishing.Status           Tenant.Preferred
 Carpet Area:2106   Bangalore:858   Furnished     : 603    Bachelors       : 771  
 Super Area :2346   Chennai  :861   Semi-Furnished:2123    Bachelors/Family:3259  
                    Delhi    :543   Unfurnished   :1726    Family          : 422  
                    Hyderabad:832                                                 
                    Kolkata  :492                                                 
                    Mumbai   :866                                                 
    Bathroom          Point.of.Contact
 Min.   :1.000   Contact Agent:1376  
 1st Qu.:1.000   Contact Owner:3076  
 Median :2.000                       
 Mean   :1.915                       
 3rd Qu.:2.000                       
 Max.   :5.000                       
```

As a result of the comprehensive data cleaning process, sparse categories were removed and the maximum values of several variables were significantly reduced: the maximum Rent dropped from 3,500,000 to 650,000, Size decreased from 8,000 to 4,500, and the number of Bathrooms was capped at 5, eliminating clearly erroneous entries. The resulting cleaned dataset consists of **4,452 observations**, which represents a robust sample size suitable for reliable statistical modeling and regression analysis. At this stage, the data has been thoroughly pre-processed and is ready for further exploration, modeling, and interpretation.

Data Visualization:

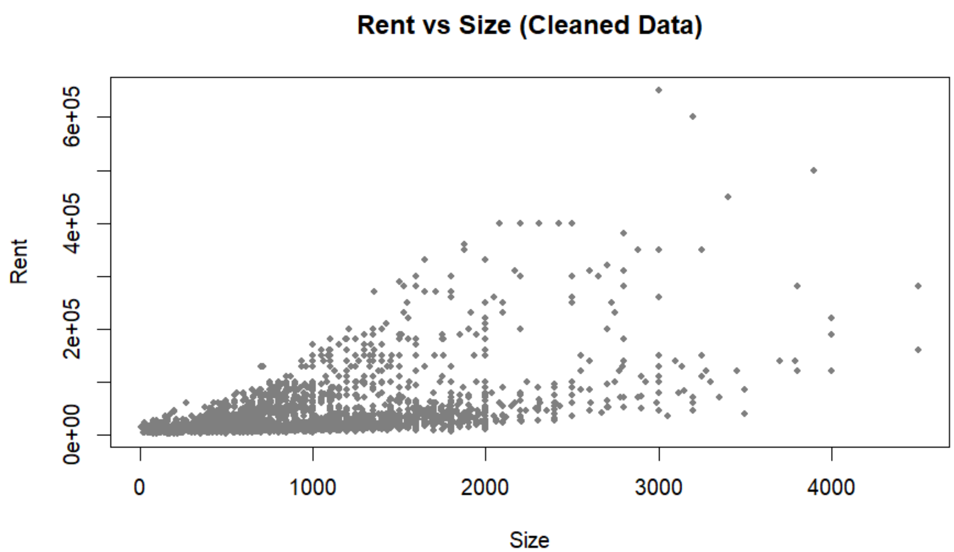


Figure 1: Scatterplot of Rent vs Size (Cleaned Data):
This scatterplot displays the relationship between property size (in square feet) and monthly rent. The **positive trend** suggests as the size of the property increases, rent tends to increase as well. This is consistent with economic intuition—larger properties generally command higher rental prices.

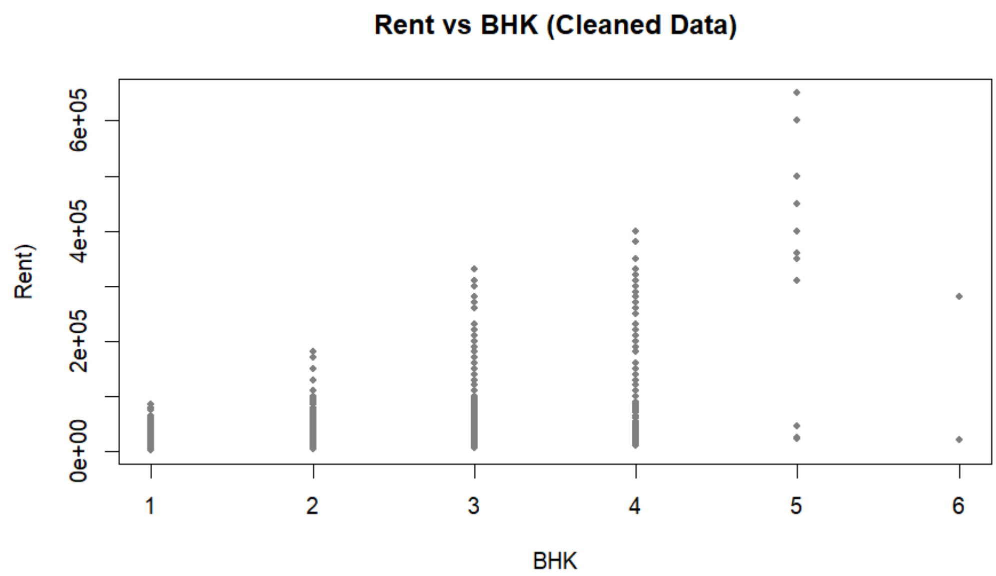


Figure 2: Scatterplot of Rent vs BHK (Cleaned Data):
This plot examines the relationship between the number of BHK (bedroom-hall-kitchen units) and the monthly rent. A general **upward trend** is observed, indicating that properties with more rooms tend to have higher rents. However, there is considerable overlap in rent values across different BHK categories, suggesting that BHK alone may not be a strong predictor of rent without accounting for other factors such as size, location, and furnishing. This supports the need for multivariate analysis.

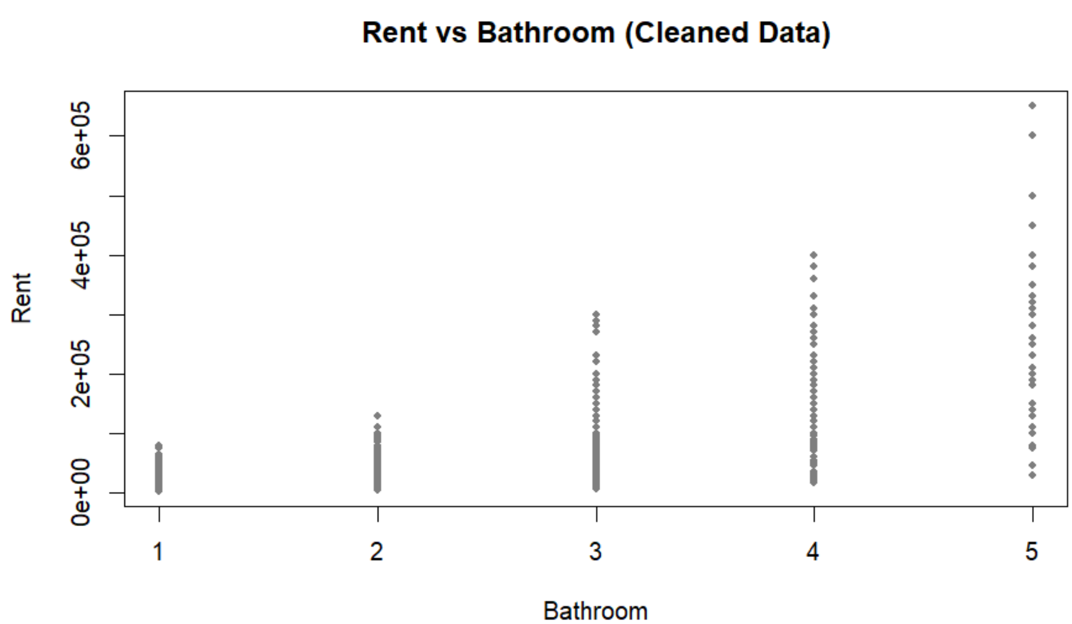


Figure 3: Scatterplot of Rent vs Bathroom (Cleaned Data):
This graph shows the distribution of rent in relation to the number of bathrooms in each property. As expected, properties with more bathrooms generally have higher rents, suggesting a positive association between amenities and pricing. However, similar to the trend observed with BHK, there's noticeable variation in rent within each bathroom category. This again indicates the importance of considering multiple factors simultaneously to explain rent variations accurately.

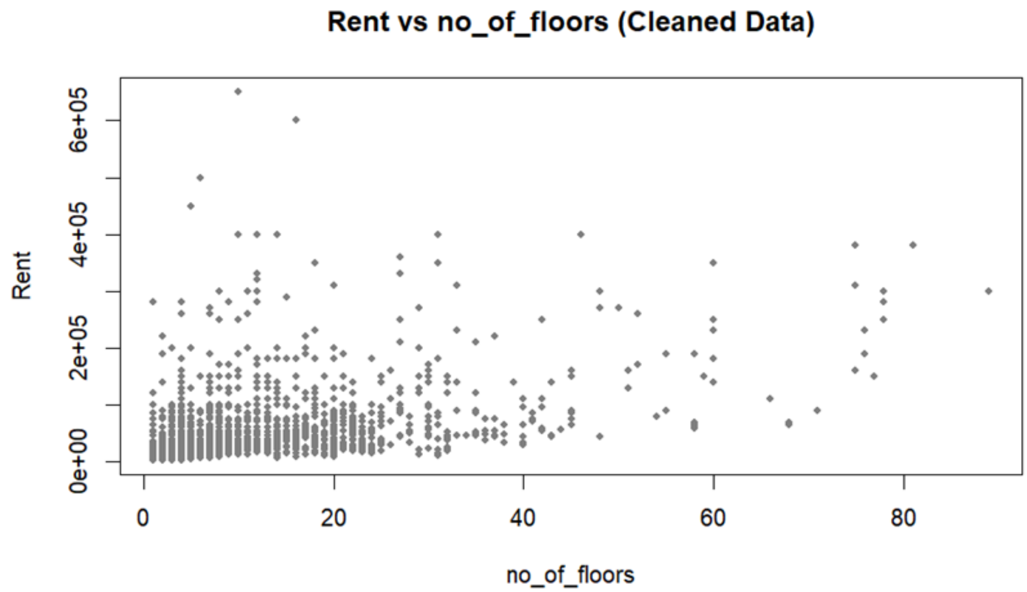


Figure 4: Scatterplot of Rent vs Number of Floors (Cleaned Data):
This scatterplot illustrates the distribution of rent in relation to the total number of floors in the building. While the relationship appears less structured compared to other variables, there is a slight tendency for higher rents in taller buildings. However, the rent values remain widely spread across different floor counts, suggesting that the total number of floors may have a more indirect or contextual influence on rent—possibly reflecting building type or urban density rather than a direct causal effect.

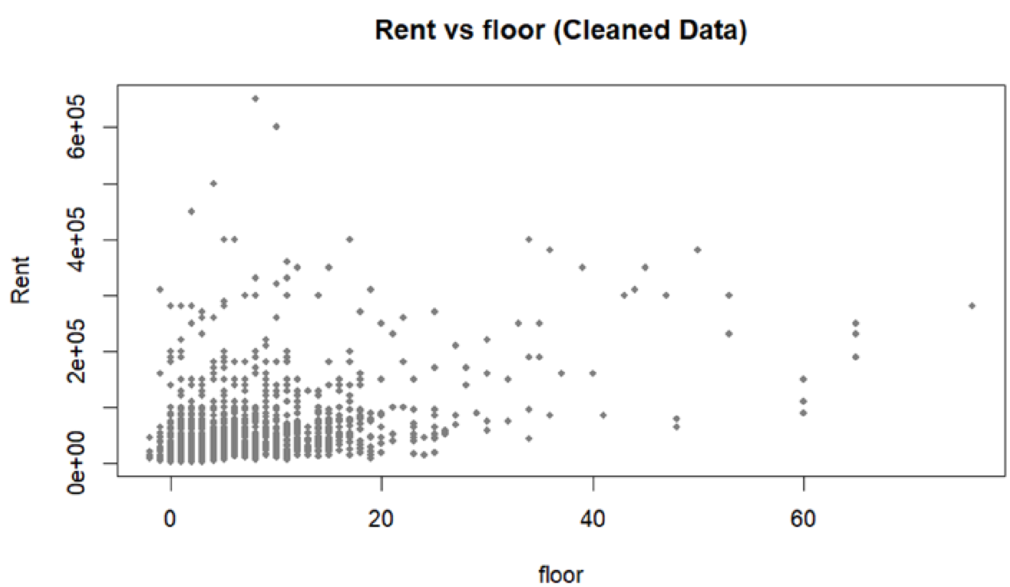


Figure 5: Scatterplot of Rent vs Floor (Cleaned Data):
This plot explores the relationship between the specific floor level a property is located on and its rent price. The distribution appears relatively dispersed, with no strong linear trend. While some higher-floor units are associated with increased rent, the overall variation in rent across floor levels suggests that the floor variable alone does not consistently explain rent differences. Its influence may depend on city-specific preferences, building type, or view-related premiums, making it more useful when considered in interaction with other variables.

For Summary for scatterplots: Exploring the Direction of Variable Relationships

The visual analysis of the scatterplots reveals several patterns in how independent variables relate to rent. There is a **direct relationship between property size and rent**—as size increases, rent tends to increase as well. This trend is expected, as larger properties offer more space and amenities, which generally command higher rental prices. Similarly, the number of **BHK units** and **bathrooms** show a **positive association with rent**, indicating that more spacious and well-equipped homes typically cost more.

For structural features like **number of floors in the building** and the **specific floor level of the unit**, the relationships are less clear. The **number of floors** shows a mild positive pattern, possibly reflecting the premium of newer or high-rise buildings in urban markets. The **floor level** variable, however, exhibits more scattered behavior, implying a weaker or context-dependent effect on rent—possibly influenced by preferences for middle floors in India due to heat or accessibility considerations.

None of the examined variables so far appears to have an **inverse** relationship with rent. Instead, most factors either positively correlate with rent or show **non-linear** or **inconclusive patterns**. This suggests that a multivariate econometric model is necessary to isolate the true effects of each variable while controlling for others.

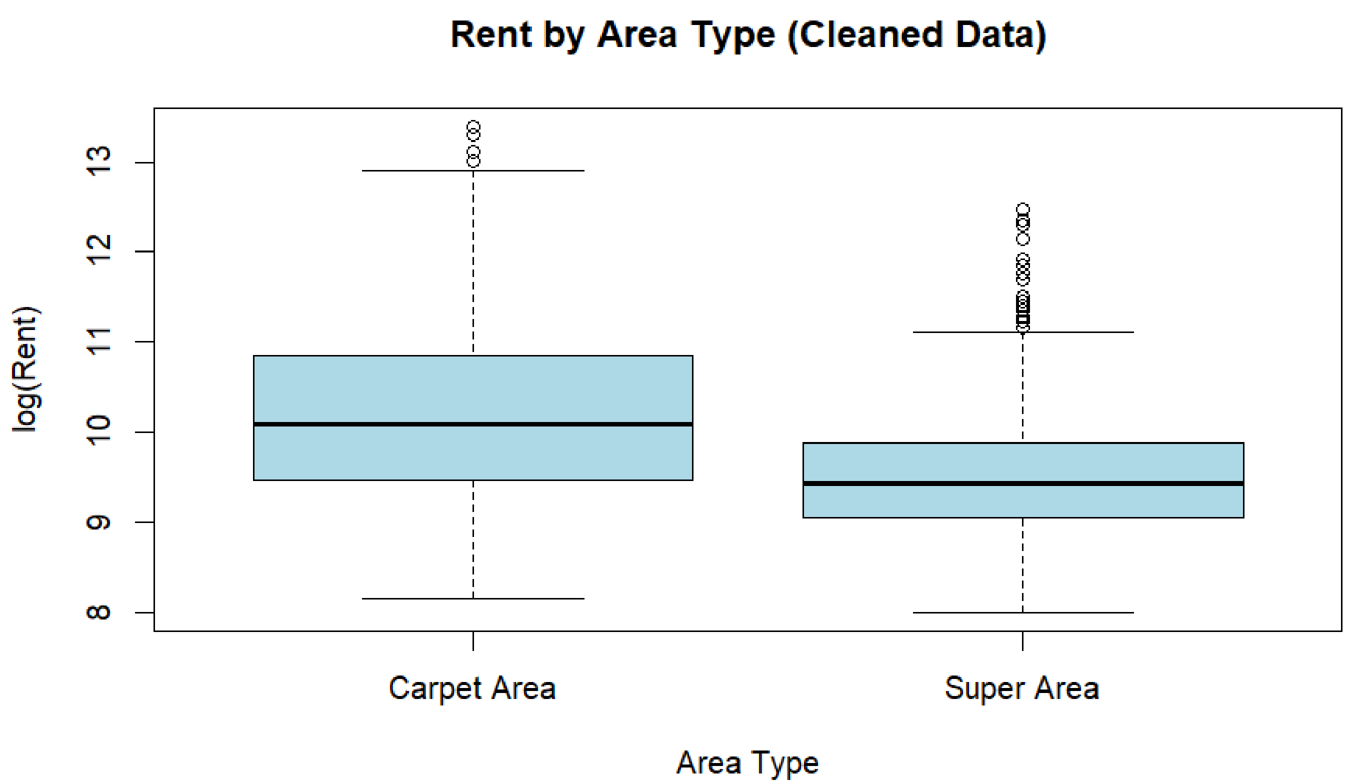


Figure 6: Boxplot of Log(Rent) by Area Type (Cleaned Data):
This boxplot compares the distribution of log-transformed rent values across two types of area measurements: Carpet Area and Super Area. The results indicate that properties measured using **Carpet Area** generally command **higher rents** than those listed by **Super Area**, with a higher median and broader upper quartile range. This may reflect more transparent or efficient pricing when actual usable space (Carpet Area) is highlighted. The boxplot also shows a greater spread and number of outliers among Carpet Area listings, suggesting a wider variation in premium offerings or market valuation.

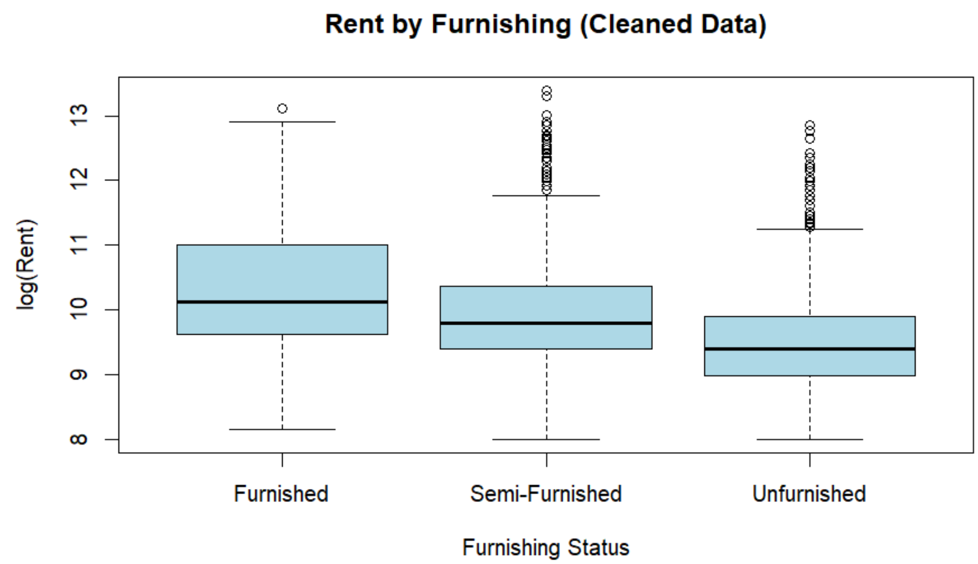


Figure 7: Boxplot of Log(Rent) by Furnishing Status (Cleaned Data):
This boxplot compares log-transformed rent values across different furnishing statuses: Furnished, Semi-Furnished, and Unfurnished. The **Furnished** category clearly shows the **highest median rent**, followed by **Semi-Furnished**, and finally **Unfurnished** properties with the lowest median. This suggests a **direct relationship** between furnishing level and rental price, likely due to the added convenience and reduced setup costs for tenants. The wider spread and higher outliers in the semi-furnished and unfurnished categories may indicate more heterogeneous pricing in those segments.

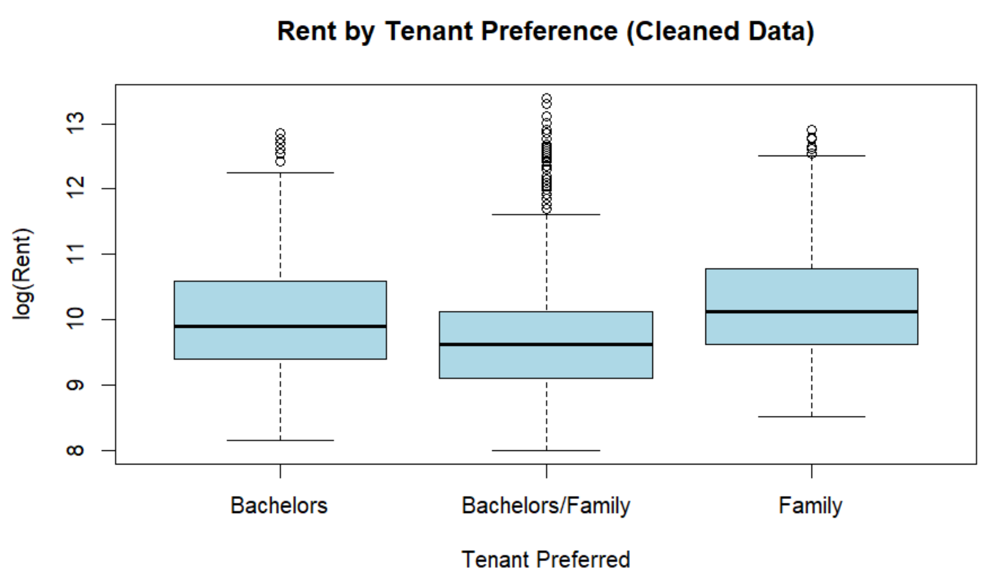


Figure 8: Boxplot of Log(Rent) by Tenant Preference (Cleaned Data):
This plot presents the distribution of log-transformed rent based on the preferred type of tenant: Bachelors, Bachelors/Family, and Family. Properties rented specifically to **Families** tend to have the **highest median rent**, while those open to both **Bachelors and Families** exhibit slightly lower medians and a broader spread. Listings targeting only **Bachelors** also show a moderately high median but with less dispersion. These differences suggest that landlords may set higher rents when targeting families, possibly due to expectations of longer tenancy and better property maintenance.

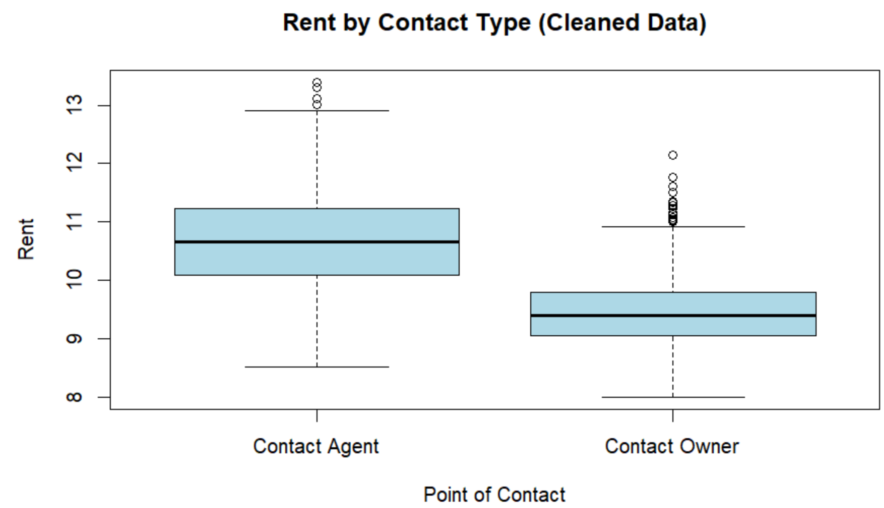


Figure 9: Boxplot of Log(Rent) by Point of Contact (Cleaned Data):
This boxplot compares log-transformed rent levels based on the point of contact for rental inquiries—either a **Contact Agent** or the **Property Owner**. The data shows that listings handled by **agents** tend to have a **higher median rent** and a broader interquartile range compared to those managed directly by owners. This may reflect the nature of listings handled by agents, which are often located in premium areas or represent higher-value properties. It also suggests that agent-mediated rentals could be associated with more formalized or commercial transactions.

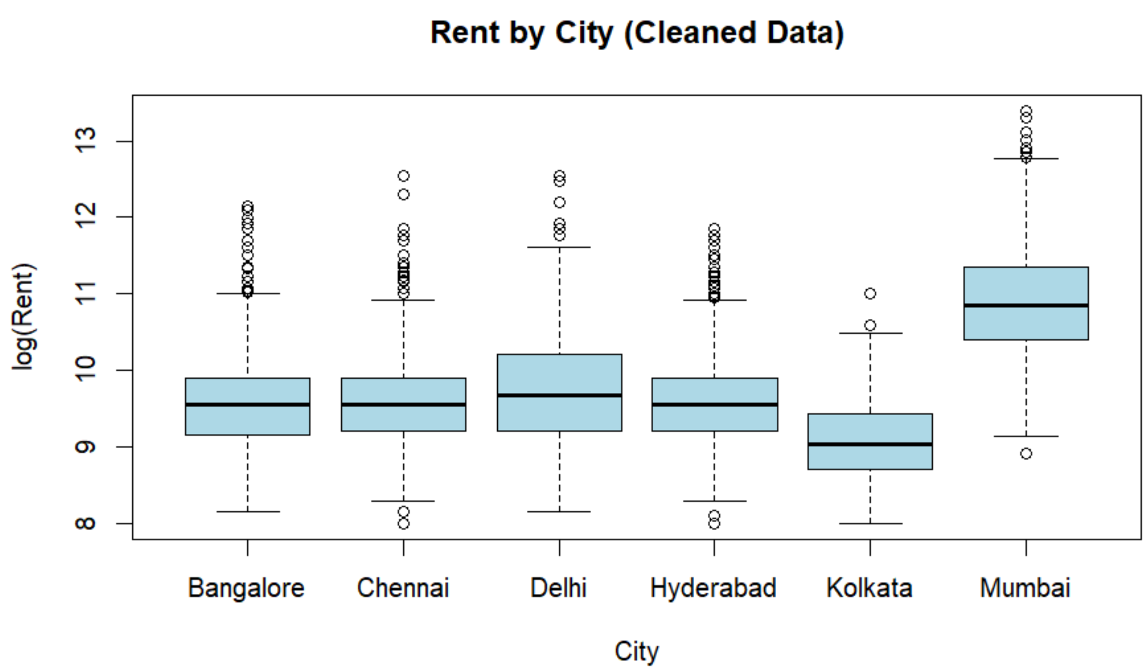


Figure 10: Boxplot of Log(Rent) by City (Cleaned Data):
This boxplot compares the distribution of log-transformed rent values across six major Indian cities. Among them, **Mumbai** clearly stands out with the **highest median rent** and widest interquartile range, indicating both higher rental costs and greater variability. **Delhi** also shows a relatively higher median compared to cities like **Kolkata**, which has the **lowest overall rent levels**. The results suggest significant regional disparities in the housing market, with metropolitan hubs like Mumbai and Delhi exhibiting more expensive rental conditions. These variations support further econometric investigation into geographic effects on rent, including rent per square meter.

Summary of Categorical Variable Analysis

The boxplots provide valuable insights into how various categorical factors influence rent levels in the dataset. **Furnishing status** shows a clear, direct relationship with rent: **furnished properties** have the highest median rent, followed by **semi-furnished**, and finally **unfurnished** units, reflecting the added value of move-in-ready housing. Similarly, **tenant preference** plays a role, with **family-targeted listings** commanding higher rents on average than those open to **bachelors** or both, possibly due to expectations of longer tenancy or better upkeep.

For **area type**, properties listed by **Carpet Area** consistently report higher rents than those listed by **Super Area**, suggesting that renters may value transparency in usable living space or that Carpet Area listings reflect higher-quality or more premium properties. Likewise, the **point of contact** also matters—properties managed by **agents** tend to have higher rents compared to those listed directly by owners, likely indicating that agents handle more commercial or upscale listings.

The most pronounced variation appears across **cities**. **Figure 10** shows a significant disparity in rent levels among the six urban centers. **Mumbai** leads with the highest median and a broad upper range, confirming its status as India's most expensive rental market. **Delhi** ranks second, while **Kolkata** records the lowest rent levels. **Bangalore**, **Chennai**, and **Hyderabad** occupy the middle ground with relatively similar distributions. These differences highlight the geographic dimension of rent determination and justify further econometric testing, particularly in analyzing how city-specific factors and interaction effects (e.g., rent per square meter, area type by city) shape the housing market.

Main effect regression model:

Initial model:

An initial ANCOVA model was conducted including all available variables to assess their joint effects on the dependent variable rent using the cleaned dataset.

```
summary(model)

Call:
lm(formula = Rent ~ Size + BHK + Bathroom + City + Area.Type +
    Furnishing.Status + floor + no_of_floors + Tenant.Preferred +
    Point.of.Contact, data = df_clean)

Residuals:
   Min     1Q Median     3Q    Max
-74087 -12911  -1137   9689 480606

Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                      -23412.68    2279.67 -10.270  < 2e-16 ***
Size                                 25.03       1.27  19.709  < 2e-16 ***
BHK                                3747.36     951.82   3.937 8.38e-05 ***
Bathroom                          10958.62     993.66  11.029  < 2e-16 ***
CityChennai                       -2654.61    1319.78  -2.011 0.044344 *
CityDelhi                          8354.55    1545.74   5.405 6.83e-08 ***
CityHyderabad                     -9797.57    1345.32  -7.283 3.85e-13 ***
CityKolkata                        1955.29    1607.61   1.216 0.223947
CityMumbai                        38763.53    1683.18  23.030  < 2e-16 ***
Area.TypeSuper Area               -2194.66    1048.78  -2.093 0.036442 *
Furnishing.StatusSemi-Furnished   -9061.36    1263.94  -7.169 8.80e-13 ***
Furnishing.StatusUnfurnished      -8658.83    1310.07  -6.609 4.31e-11 ***
floor                               947.53     152.20   6.225 5.25e-10 ***
no_of_floors                        343.01     103.47   3.315 0.000923 ***
Tenant.PreferredBachelors/Family   3470.28    1178.60   2.944 0.003252 **
Tenant.PreferredFamily            -3425.17    1679.17  -2.040 0.041430 *
Point.of.ContactContact Owner     -5636.92    1270.20  -4.438 9.31e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 26890 on 4435 degrees of freedom
Multiple R-squared:  0.638,	Adjusted R-squared:  0.6367
F-statistic: 488.5 on 16 and 4435 DF,  p-value: < 2.2e-16
```

- The **overall F-statistic** of the model is **488.5** with a **p-value < 2.2e-16**, which is **highly significant**. This indicates that **at least one** of the predictors in the model has a statistically significant relationship with `Rent`. In other words, the model **as a whole** explains a significant portion of the variance in the target variable.

- Adjusted R-squared: 0.6367, This means that approximately **63.6%** of the variance in rental prices (`Rent`) is explained by the model. Adjusted R-squared is very close to the R-squared, indicating that **most included predictors are contributing** and not just adding noise.

- In a proper linear regression, standardized residuals should lie roughly between -3 and +3, indicating normality. However, here the residuals range from about -74,087 to +480,606 (raw values), showing they are not normally distributed. This can undermine statistical tests like t-tests and F-tests. The issue likely stems from skewness in Rent, which can often be improved by applying a log transformation to the dependent variable.

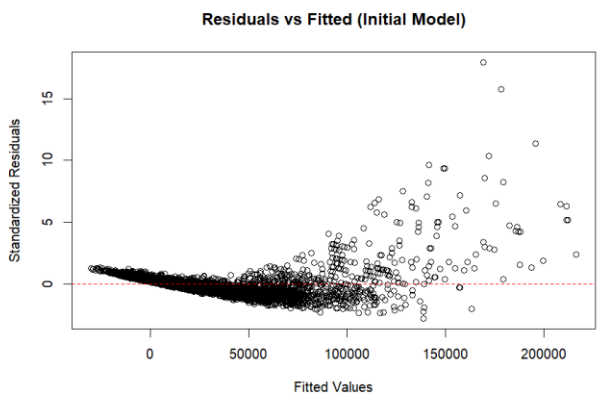
Test multicollinearity for initial model:

```
vif(model)
                      GVIF Df GVIF^(1/(2*Df))
Size              2.962901  1        1.721308
BHK               3.464946  1        1.861437
Bathroom          3.944198  1        1.986000
City              2.839725  5        1.110012
Area.Type         1.687985  1        1.299225
Furnishing.Status 1.164703  2        1.038852
floor             4.436573  1        2.106317
no_of_floors      5.144049  1        2.268050
Tenant.Preferred  1.270554  2        1.061691
Point.of.Contact  2.121109  1        1.456403
```

To assess multicollinearity, **Variance Inflation Factor (VIF)** values were examined for all predictors. All VIFs are **below 10**, with the highest being around **5.14** for no_of_floors. Since none of the values exceed the commonly accepted threshold of **10**, we can conclude that **multicollinearity is not a concern** in this model.

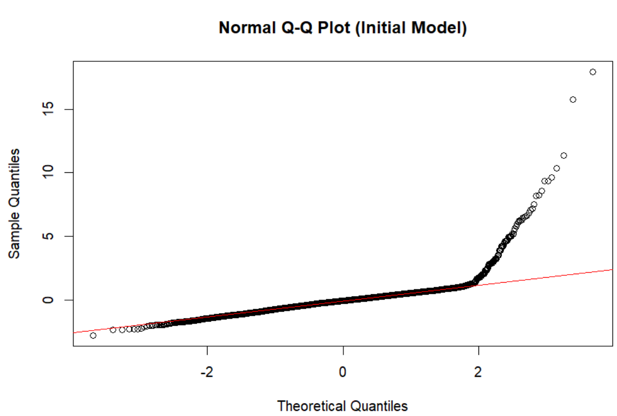
Check assumptions about error term in initial model:

1- The variability in random error is constant (Homoscedasticity) and the Expected value of the error term equals zero, $E(\epsilon_i) = 0$.



The residual plot reveals clear signs of **heteroscedasticity**, as the variance of the residuals increases with higher fitted values. Additionally, the residuals are **not randomly scattered** around zero; instead, they follow a visible pattern. This indicates a violation of the assumption of linear regression.

2. check assumption normality of the error term for initial model:




```
shapiro.test(residuals(model))

        Shapiro-Wilk normality test

data:  residuals(model)
W = 0.69268, p-value < 2.2e-16

> ad.test(residuals(model))

        Anderson-Darling normality test

data:  residuals(model)
A = 179.41, p-value < 2.2e-16
```

To test the normality of residuals, both the **Shapiro-Wilk** and **Anderson-Darling** tests were performed. In both cases, the p-values are **extremely small**, far below the common alpha level (e.g., 0.05), indicating that we **reject the null hypothesis** of normality. This confirms that the residuals are **not normally distributed**, which supports what was observed visually in Q-Q plot as there is a large deviations from 45-degree line at large values.

3- check Are the errors independent (no auto -correlated between error) using, $\boldsymbol{cov}\ (\boldsymbol{\epsilon i}, \boldsymbol{\epsilon j}) = \mathbf{0}$:

```
durbinWatsonTest(model)
 lag Autocorrelation D-W Statistic p-valu
e
   1   -0.0008150032      2.001574   0.93
8
 Alternative hypothesis: rho != 0
```

```
resettest(model, power = 2:3, type = "fit
ted")

        RESET test

data:  model
RESET = 3070.9, df1 = 2, df2 = 4433, p-va
lue < 2.2e-16
```

Since the Durbin-Watson statistic is very close to 2 and the p-value is large, we fail to reject the null hypothesis of no autocorrelation. This means **there is no significant evidence of autocorrelation in the residuals at lag 1**, indicating that the residuals are likely independent.

Since the p-value for reset test is far below the typical 0.05 threshold, we **reject the null hypothesis** that the **model is correctly specified.** This suggests that the current model **may suffer from misspecification**, possibly due to incorrect functional forms, or other structural issues.

To address the issues identified—non-normality of residuals, heteroscedasticity, and potential model misspecification—a logarithmic transformation of the dependent variable, Rent, will be applied. Although the dataset has been cleaned to remove obvious outliers, some extreme values in Rent remain because they reflect real market conditions rather than errors. Applying log(Rent) helps stabilize variance, reduce the effect of these genuine high values, improve the normality of residuals, and enhance the overall model fit. This transformation supports meeting classical linear regression assumptions, thereby improving the reliability of inference and prediction.

Log – Transformed Model:

```
summary(log_model)

Call:
lm(formula = log(Rent) ~ Size + BHK + Bathroom + City + Area.Type +
    Furnishing.Status + floor + no_of_floors + Tenant.Preferred +
    Point.of.Contact, data = df_clean)

Residuals:
     Min       1Q   Median       3Q      Max
-0.94580 -0.21747 -0.01334  0.20547  1.04987

Coefficients:
                                   Estimate Std. Error t value Pr(>|t|)
(Intercept)                       8.973e+00  2.682e-02 334.529  < 2e-16 ***
Size                              4.617e-04  1.494e-05  30.897  < 2e-16 ***
BHK                               2.049e-01  1.120e-02  18.294  < 2e-16 ***
Bathroom                          1.549e-01  1.169e-02  13.248  < 2e-16 ***
CityChennai                      -2.782e-02  1.553e-02  -1.791 0.073311 .
CityDelhi                         1.844e-01  1.819e-02  10.142  < 2e-16 ***
CityHyderabad                    -1.347e-01  1.583e-02  -8.513  < 2e-16 ***
CityKolkata                      -2.811e-01  1.892e-02 -14.861  < 2e-16 ***
CityMumbai                        9.590e-01  1.980e-02  48.426  < 2e-16 ***
Area.TypeSuper Area              -4.796e-02  1.234e-02  -3.886 0.000103 ***
```

```
Furnishing.StatusSemi-Furnished  -1.796e-01  1.487e-02 -12.078  < 2e-16 ***
Furnishing.StatusUnfurnished     -2.939e-01  1.541e-02 -19.068  < 2e-16 ***
floor                             2.460e-03  1.791e-03   1.374 0.169556
no_of_floors                      4.673e-03  1.217e-03   3.838 0.000126 ***
Tenant.PreferredBachelors/Family -2.938e-02  1.387e-02  -2.119 0.034162 *
Tenant.PreferredFamily           -8.471e-02  1.976e-02  -4.288 1.84e-05 ***
Point.of.ContactContact Owner    -3.256e-01  1.495e-02 -21.788  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3164 on 4435 degrees of freedom
Multiple R-squared:  0.8688,   Adjusted R-squared:  0.8683
F-statistic:  1836 on 16 and 4435 DF,  p-value: < 2.2e-16
```

- **Residuals:** The residuals of the log-transformed model range between approximately -0.95 and 1.05, which falls well within the acceptable range of -3 to 3. This indicates that there are no significant outliers affecting the rental rates in the transformed model.
- **Model significance:** The overall model is highly significant, with a p-value < 2.2e-16, which is far below the 0.05 significance threshold.
- **Adjusted R-squared:** The log(Rent) model explains about 86.83% of the variance in the dependent variable, which is a notable improvement compared to the original model (Adjusted R-squared ≈ 0.6367). This shows that the log transformation provides a better fit and explains more variability in rent prices.

Comparison with Initial model:

Metric	Original Model	Log-Transformed Model
Residual range	-74,087 to +480,606	-0.94580 to +1.04987
Adjusted R-squared	0.6367	0.8683
Model p-value	< 2.2e-16	< 2.2e-16
Residual outliers	Present (large residuals)	None (within -3 to 3 range)
Model fit quality	Moderate	Strong

The log transformation of Rent greatly improves the model assumptions.

Test multicollinearity for log model:

```
vif(log_model)
                     GVIF Df GVIF^(1/(2*Df))
Size             2.962901  1        1.721308
BHK              3.464946  1        1.861437
Bathroom         3.944198  1        1.986000
City             2.839725  5        1.110012
Area.Type        1.687985  1        1.299225
Furnishing.Status 1.164703  2        1.038852
floor            4.436573  1        2.106317
no_of_floors     5.144049  1        2.268050
Tenant.Preferred 1.270554  2        1.061691
Point.of.Contact 2.121109  1        1.456403
```

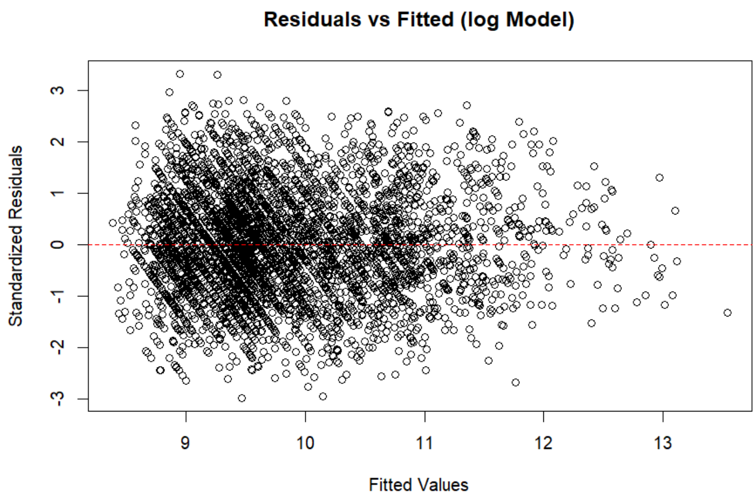
To assess multicollinearity, Variance Inflation Factor (VIF) values were examined for all predictors. All VIFs are below 10, with the highest being around 5.14 for no_of_floors. **Since none of the values exceed the commonly accepted threshold of 10, we can conclude that multicollinearity is not a concern in this model.**

Testing the Significance of Individual Predictors:

- **H_0: β0 = 0 Vs H_1: β0 ≠ 0**
  P_value = < 2e-16 < 0.05
  Decision: Reject H_0. The intercept is statistically significant, indicating a meaningful baseline rent on the log scale.
- **H_0: β1 = 0 Vs H_1: β1 ≠ 0 (Size)**
  P_value = < 2e-16 < 0.05
  Decision: Reject H_0. Size is statistically significant, indicating larger houses have higher rental prices.
- **H_0: β2 = 0 Vs H_1: β2 ≠ 0 (BHK)**
  P_value = < 2e-16 < 0.05
  Decision: Reject H_0. BHK is significant, more bedrooms increase rent.
- **H_0: β3 = 0 Vs H_1: β3 ≠ 0 (Bathroom)**
  P_value = < 2e-16 < 0.05
  Decision: Reject H_0. Bathrooms significantly increase rent.
- **H_0: β4 = 0 Vs H_1: β4 ≠ 0 (CityChennai)**
  P_value = 0.073311 > 0.05
  Decision: Fail to reject H_0. Chennai is not statistically significant.
- **H_0: β5 = 0 Vs H_1: β5 ≠ 0 (CityDelhi)**
  P_value = < 2e-16 < 0.05
  Decision: Reject H_0. Delhi significantly increases rent.
- **H_0: β6 = 0 Vs H_1: β6 ≠ 0 (CityHyderabad)**
  P_value = < 2e-16 < 0.05
  Decision: Reject H_0. Hyderabad significantly decreases rent.
- **H_0: β7 = 0 Vs H_1: β7 ≠ 0 (CityKolkata)**
  P_value = < 2e-16 < 0.05
  Decision: Reject H_0. Kolkata significantly decreases rent.
- **H_0: β8 = 0 Vs H_1: β8 ≠ 0 (CityMumbai)**
  P_value = < 2e-16 < 0.05
  Decision: Reject H_0. Mumbai strongly increases rent.
- **H_0: β9 = 0 Vs H_1: β9 ≠ 0 (Area.TypeSuper Area)**
  P_value = 0.000103 < 0.05
  Decision: Reject H_0. Super Area type significantly decreases rent.
- **H_0: β10 = 0 Vs H_1: β10 ≠ 0 (Furnishing.StatusSemi-Furnished)**
  P_value = < 2e-16 < 0.05
  Decision: Reject H_0. Semi-furnished status significantly decreases rent.
- **H_0: β11 = 0 Vs H_1: β11 ≠ 0 (Furnishing.StatusUnfurnished)**
  P_value = < 2e-16 < 0.05
  Decision: Reject H_0. Unfurnished status significantly decreases rent.
- **H_0: β12 = 0 Vs H_1: β12 ≠ 0 (floor)**
  P_value = 0.169556 > 0.05
  Decision: Fail to reject H_0. Floor level is not significant.
- **H_0: β13 = 0 Vs H_1: β13 ≠ 0 (no_of_floors)**
  P_value = 0.000126 < 0.05
  Decision: Reject H_0. Number of floors significantly increases rent.
- **H_0: β14 = 0 Vs H_1: β14 ≠ 0 (Tenant.PreferredBachelors/Family)**
  P_value = 0.034162 < 0.05
  Decision: Reject H_0. Tenant preference Bachelors/Family significantly affects rent.
- **H_0: β15 = 0 Vs H_1: β15 ≠ 0 (Tenant.PreferredFamily)**
  P_value = 1.84e-05 < 0.05
  Decision: Reject H_0. Tenant preference for Family significantly decreases rent.
- **H_0: β16 = 0 Vs H_1: β16 ≠ 0 (Point.of.ContactContact Owner)**
  P_value = < 2e-16 < 0.05
  Decision: Reject H_0. Contacting the owner directly significantly decreases rent.

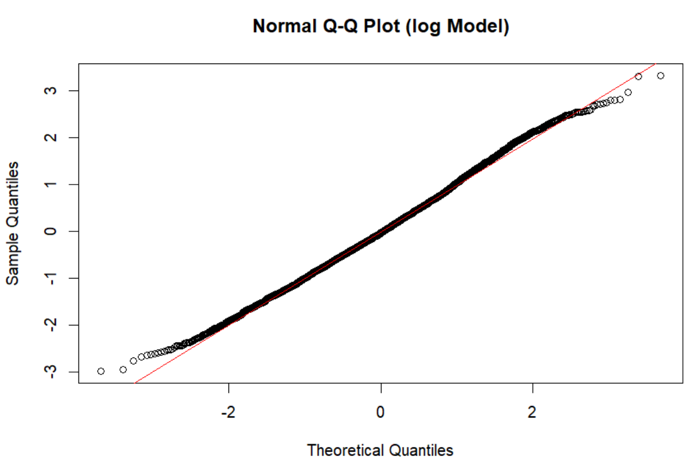
Check assumptions about error term in initial model:

1- The variability in random error is constant (Homoscedasticity) and the Expected value of the error term equals zero, $\boldsymbol{E(\epsilon_i) = 0}$



The residuals vs. fitted plot showed a **mostly random scatter** around zero. However, a **slight funnel shape** (mild heteroscedasticity) was observed, likely due to the influence of the **size variable**. While this minor pattern doesn't severely violate model assumptions, it may warrant further investigations such as transforming the size variable or using robust standard errors—to ensure more reliable inference.

2- check assumption normality of the error term for initial model:

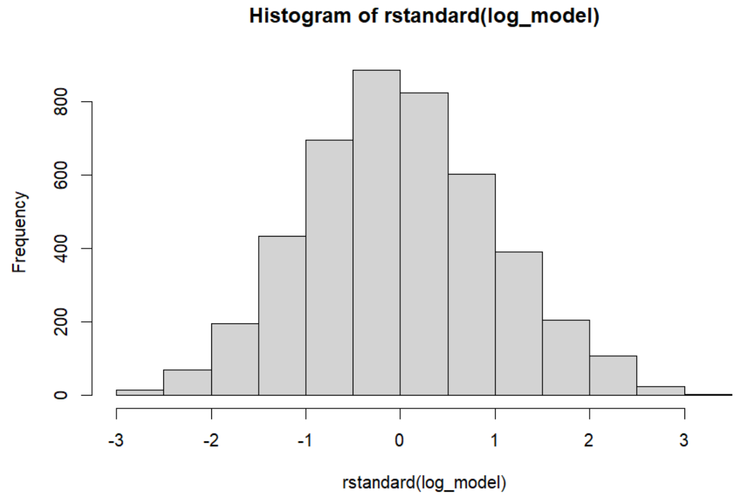



```
shapiro.test(residuals(log_model))

        Shapiro-Wilk normality test

data:  residuals(log_model)
W = 0.99738, p-value = 6.202e-07

> ad.test(residuals(log_model))

        Anderson-Darling normality test

data:  residuals(log_model)
A = 2.6497, p-value = 1.116e-06
```

(H_0): Residuals follows a normal distribution.

(H_1): Residuals does not follow a normal distribution.

Although the Q-Q plot shows residuals closely following the normal line and the Shapiro-Wilk statistic is near 1, the very small p-values from both Shapiro-Wilk and Anderson-Darling tests lead us to reject the null hypothesis of normality. This creates a conflict between the visual evidence and the formal tests, likely due to the large sample size making the tests highly sensitive. Hence, further investigation like examining the residuals' histogram is needed.



The graph is symmetric, which provides evidence that the residuals follow a normal distribution. Therefore, we can **disregard the conflicting results** from the **Shapiro-Wilk** and **Anderson-Darling** normality tests. Additionally, the residuals lie within the range of **-3 to 3**, **which further supports the assumption of normality.**

3- check Are the errors independent (no auto -correlated between error) using, $cov(\epsilon i, \epsilon j) = 0$:

```
durbinWatsonTest(log_model)
 lag Autocorrelation D-W Statistic p-value
   1     0.009320313      1.981283   0.472
 Alternative hypothesis: rho != 0
```

H_0: No first-order autocorrelation in residuals ($\rho_1 = 0$)
H_1: Presence of first-order autocorrelation ($\rho_1 \neq 0$)

The Durbin-Watson statistic is approximately 1.98, very close to 2, indicating no autocorrelation in the residual. The p-value (0.472) is greater than 0.05, so we fail to reject the null hypothesis. **This confirms residuals are likely independent with no significant autocorrelation.**

```
resettest(log_model, power = 2:3, type = "fitted")
        RESET test

data:  log_model
RESET = 4.0761, df1 = 2, df2 = 4433, p-value = 0.01704
```

(H_0): the model is correctly specified
(H_1): the model is misspecified.

The RESET test produced a test statistic of **4.0761** with a **p-value of 0.01704**. At the **5% significance level**, this p-value leads to **rejection of the null hypothesis**, suggesting potential issues with the model specification. While the p-value is **greater than 0.01**, it remains **below 0.05**, indicating **moderate but statistically significant evidence** of misspecification.

This still represents a **substantial improvement** over the initial (non-log-transformed) model, which showed **severe misspecification** with a RESET statistic of **3070.9** and a **p-value < 2.2e-16**, strongly rejecting the null hypothesis.

Test	Initial Model	Log-Transformed Model	Interpretation
Shapiro-Wilk	W = 0.6927, p < 2.2e-16	W = 0.9974, p = 6.202e-07	Strong improvement in residual normality after log transformation.
Anderson-Darling	A = 179.41, p < 2.2e-16	A = 2.6497, p = 1.116e-06	reduction in non-normality; distribution more normal after log transform.
Durbin-Watson	DW = 2.0016, p = 0.938	DW = 1.9813, p = 0.472	No autocorrelation detected in either model.
RESET Test	F = 3070.9, p < 2.2e-16	F = 4.0761, p = 0.01704	Strong misspecification in initial model; mostly resolved in log model

The log-transformed model demonstrates clear statistical superiority over the initial model across all diagnostic measures. However, although the log transformation has addressed most violations of linear regression assumptions, a slight funnel shape remains in the residual plot, suggesting heteroscedasticity concern. Moreover, the scatter plot of Rent against Size shows a funnel out-shaped pattern, where the variability in Rent increases with Size. So, we will try log transformation of the "Size" variable will be considered in the next iteration.

Log – Log model:

```
summary(log_log_model)
Call:
lm(formula = log(Rent) ~ log(Size) + floor + BHK + Bathroom +
    City + Area.Type + Furnishing.Status + no_of_floors + Tenant.Preferred +
    Point.of.Contact, data = df_clean)
Residuals:
     Min       1Q   Median       3Q      Max
-1.03913 -0.22717 -0.00883  0.21624  1.08092
Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                       7.775798   0.067063 115.948  < 2e-16 ***
log(Size)                         0.212964   0.010370  20.537  < 2e-16 ***
floor                             0.002189   0.001887   1.161  0.24588
BHK                               0.256848   0.011556  22.226  < 2e-16 ***
Bathroom                          0.234274   0.011757  19.927  < 2e-16 ***
CityChennai                      -0.044414   0.016346  -2.717  0.00661 **
CityDelhi                         0.195522   0.019745   9.903  < 2e-16 ***
CityHyderabad                    -0.114656   0.016663  -6.881 6.78e-12 ***
CityKolkata                      -0.288644   0.019929 -14.484  < 2e-16 ***
CityMumbai                        0.874209   0.020471  42.704  < 2e-16 ***
Area.TypeSuper Area              -0.028061   0.012969  -2.164  0.03054 *
Furnishing.StatusSemi-Furnished  -0.190314   0.015660 -12.153  < 2e-16 ***
Furnishing.StatusUnfurnished     -0.311492   0.016217 -19.208  < 2e-16 ***
no_of_floors                      0.005186   0.001282   4.044 5.34e-05 ***
Tenant.PreferredBachelors/Family -0.035932   0.014609  -2.460  0.01395 *
Tenant.PreferredFamily           -0.091812   0.020810  -4.412 1.05e-05 ***
Point.of.ContactContact Owner    -0.364230   0.015636 -23.294  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3333 on 4435 degrees of freedom
Multiple R-squared:  0.8544,	Adjusted R-squared:  0.8539
F-statistic:  1627 on 16 and 4435 DF,  p-value: < 2.2e-16
```

- The log-log-transformed model, which replaces *Size* with *log(Size)*, maintains strong overall significance with a p-value less than 2.2e-16.
- The model shows good explanatory power with an Adjusted R^2 of 0.8539, indicating that about 85.39% of the variation in log(Rent) is explained by the predictors. However, this represents a slight decrease in R^2 compared to the previous log model.
- The residuals of the reduced log-transformed model fall within the normal range of -3 to 3, indicating no extreme outliers and suggesting a good model fit.

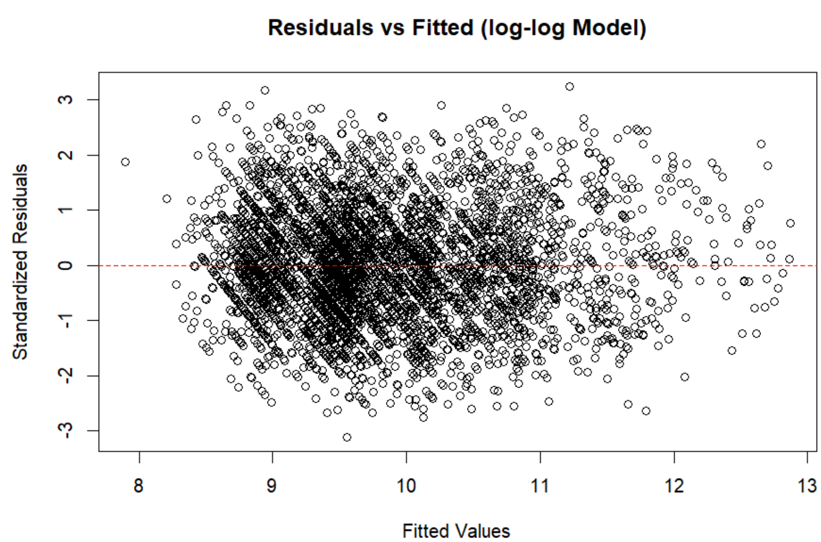
Test multicollinearity for log-log model:

```
vif(log_log_model)
                      GVIF Df GVIF^(1/(2*Df))
log(Size)         2.031513  1        1.425312
BHK               3.324561  1        1.823338
Bathroom          3.594023  1        1.895791
City              2.913593  5        1.112866
Area.Type         1.680075  1        1.296177
Furnishing.Status 1.162830  2        1.038434
floor             4.436601  1        2.106324
no_of_floors      5.142440  1        2.267695
Tenant.Preferred  1.270927  2        1.061769
Point.of.Contact  2.092195  1        1.446442
```

To assess multicollinearity, Variance Inflation Factor (VIF) values were examined for all predictors. All VIFs are below 10, with the highest being around 5.14 for no_of_floors. **Since none of the values exceed the commonly accepted threshold of 10, we can conclude that multicollinearity is not a concern in this model.**

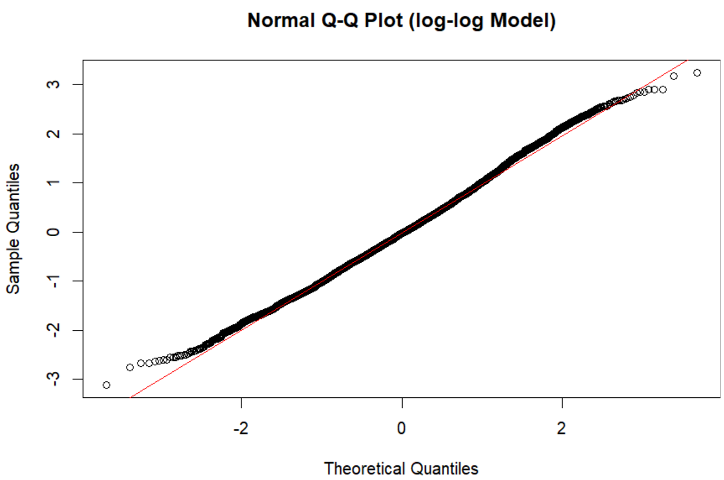
Check assumptions about error term in initial model:

1- The variability in random error is constant (Homoscedasticity) and the Expected value of the error term equals zero, $\boldsymbol{E(\epsilon_i) = 0}$



residuals plot of the reduced model shows a clear improvement compared to the initial and log-transformed models. The residuals are randomly scattered around zero with no evident pattern, indicating that the assumption of **constant variance (homoscedasticity)** holds. There is **no visible funnel shape**, which was previously noticeable in the log-transformed model. This suggests that applying log (Size) has successfully corrected the issue of non-constant variance, leading to a more reliable and well-behaved model.

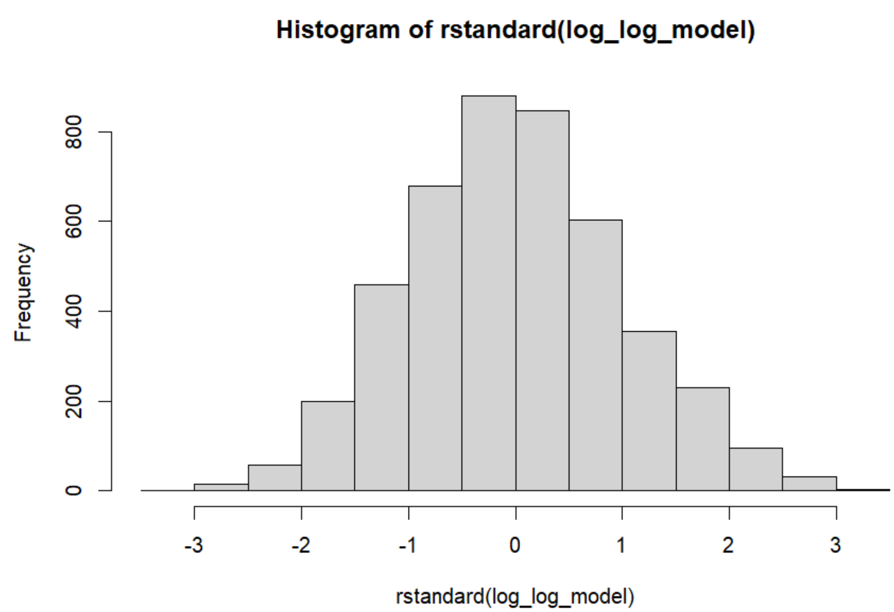
2- check assumption normality of the error term for initial model:




```
> shapiro.test(residuals(log_log_model))

        Shapiro-Wilk normality test

data:  residuals(log_log_model)
W = 0.99695, p-value = 7.582e-08

> ad.test(residuals(log_log_model))

        Anderson-Darling normality test

data:  residuals(log_log_model)
A = 2.9939, p-value = 1.621e-07
```

Both the log-transformed model (log_model) and the double log-transformed model (log_log_model) show similar results in testing the normality of residuals. The Shapiro-Wilk test for the log_log_model yielded a statistic of 0.99695 with a p-value of 7.582e-08, which is slightly lower than the log_model's 0.99738 and p-value of 6.202e-07. Likewise, the Anderson-Darling test statistic increased slightly from 2.6497 in the log_model to 2.9939 in the log_log_model, both with highly significant p-values.

Despite these statistical tests rejecting normality for both models, the Q-Q plots for the residuals of both models are very similar, showing points closely aligned with the 45-degree line. This indicates that although the tests detect minor deviations from normality, visually the residuals appear similarly well-behaved in both models.



Although the Shapiro-Wilk ($W = 0.99695$, $p < 0.001$) and Anderson-Darling ($A = 2.9939$, $p < 0.001$) tests reject normality in the residuals of the log-log model, the histogram of residuals closely resembles that of the initial log-linear model. Visually, both distributions appear approximately symmetric and bell-shaped. This consistency, along with similar Q-Q plots, suggests that the residuals are reasonably normal in both models despite statistical test sensitivities to minor deviations in large samples.

3- check Are the errors independent (no auto -correlated between error) using, $\boldsymbol{cov}\ (\boldsymbol{\epsilon i}, \boldsymbol{\epsilon j}) = \mathbf{0}$:

```
durbinWatsonTest(log_log_model)
 lag Autocorrelation D-W Statistic p-value
   1      0.02123465      1.957489   0.142
 Alternative hypothesis: rho != 0
```

H_0: No first-order autocorrelation in residuals ($\rho_1 = 0$)
H_1: Presence of first-order autocorrelation ($\rho_1 \neq 0$)

Both models yield **Durbin-Watson statistics close to 2**, indicating **no significant first-order autocorrelation** in the residuals. Although the DW statistic slightly decreased in the log-log model, the **p-value (0.142)** still exceeds the 5% significance level. Hence, we **fail to reject the null hypothesis** of no autocorrelation in both models. This supports the reliability of the regression assumptions in both cases.

```
resettest(log_log_model, power = 2:3, type = "fitted")

	RESET test

data:  log_log_model
RESET = 30.506, df1 = 2, df2 = 4433, p-value = 6.948e-14
```

In the **log model**, the RESET test gave a **moderate p-value (0.017)**, just below 5%, suggesting **some indication** of model misspecification. In contrast, the **log-log model** shows a **very large RESET statistic (30.506)** and a **highly significant p-value (< 0.000001)**, strongly indicating **serious model misspecification**—perhaps due to incorrect functional form or missing variables after transforming Size to log(Size).

Metric / Test	Log Model	Log-Log Model	Interpretation
Adjusted R^2	0.8544	0.8539	Almost identical; both models have very strong explanatory power
Residual Range	-1.04 to 1.08	-1.04 to 1.08	Same spread; residuals behave similarly in both models
Shapiro-Wilk (W / p-value)	W = 0.99738 / p = 6.2e-07	W = 0.99695 / p = 7.58e-08	Both suggest non-normal residuals, despite W values being close to 1
Anderson-Darling (A / p-value)	A = 2.6497 / p = 1.12e-06	A = 2.9939 / p = 1.62e-07	Slightly worse residual normality in the log-log model
Durbin-Watson Statistic / p-value	1.981 / p = 0.472	1.957 / p = 0.142	No autocorrelation in either model; log model shows slightly more stability
RESET Test (F / p-value)	F = 4.076 / p = 0.01704	F = 30.506 / p = 6.948e-14	Log model shows moderate misspecification significant at 5% but not 1%; log-log is clearly misspecified at all levels
Q-Q Plot & Histogram	Approximately normal	Very similar	No notable visual differences in residual distribution

While the **log-log model** slightly improves **heteroscedasticity**—as observed in the residuals' plot with a more constant variance—it comes at the cost of **increased model misspecification**, as indicated by the significantly lower p-value in the RESET test ($p < 0.001$). Therefore, despite some gain in residual behavior, the log-log specification **raises concerns about omitted variable bias or incorrect functional form**. To objectively compare both models and decide which to retain, **information criteria** such as **AIC** and **BIC** will be used as the next step.

```
> AIC(log_log_model, log_model)
              df      AIC
log_log_model 18 2870.635
log_model     18 2407.170
> BIC(log_log_model, log_model)
              df      BIC
log_log_model 18 2985.855
log_model     18 2522.390
```

Both AIC and BIC are significantly lower for the log-linear model, indicating that it provides a better balance between model fit and complexity. These further supports retaining the log-linear model over the log-log specification, despite the latter's slight improvement in heteroscedasticity.

Given that the variable floor was statistically insignificant in the log-linear model (p-value =0.169556), the next step is to build a reduced log-linear model excluding this predictor. This refinement aims to simplify the model without compromising its explanatory power or fit.

Reduced log model:

```
summary(reduced_log_model)

Call:
lm(formula = log(Rent) ~ Size + BHK + Bathroom + City + Area.Type +
    Furnishing.Status + no_of_floors + Tenant.Preferred + Point.of.Contact,
    data = df_clean)

Residuals:
     Min       1Q   Median       3Q      Max
-0.94694 -0.21854 -0.01363  0.20708  1.04700

Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                      8.971e+00  2.679e-02 334.902  < 2e-16 ***
Size                             4.614e-04  1.494e-05  30.875  < 2e-16 ***
BHK                              2.047e-01  1.120e-02  18.281  < 2e-16 ***
Bathroom                         1.556e-01  1.168e-02  13.324  < 2e-16 ***
CityChennai                     -2.765e-02  1.553e-02  -1.780   0.0751 .
CityDelhi                        1.849e-01  1.819e-02  10.168  < 2e-16 ***
CityHyderabad                   -1.346e-01  1.583e-02  -8.501  < 2e-16 ***
CityKolkata                     -2.812e-01  1.892e-02 -14.862  < 2e-16 ***
CityMumbai                       9.592e-01  1.981e-02  48.429  < 2e-16 ***
Area.TypeSuper Area             -4.815e-02  1.234e-02  -3.902 9.67e-05 ***
Furnishing.StatusSemi-Furnished -1.795e-01  1.487e-02 -12.072  < 2e-16 ***
Furnishing.StatusUnfurnished    -2.941e-01  1.542e-02 -19.082  < 2e-16 ***
no_of_floors                     6.030e-03  7.114e-04   8.477  < 2e-16 ***
Tenant.PreferredBachelors/Family -2.954e-02  1.387e-02  -2.130   0.0332 *
Tenant.PreferredFamily          -8.471e-02  1.976e-02  -4.287 1.85e-05 ***
Point.of.ContactContact Owner   -3.250e-01  1.494e-02 -21.755  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3164 on 4436 degrees of freedom
Multiple R-squared:  0.8688,	Adjusted R-squared:  0.8683
F-statistic:  1958 on 15 and 4436 DF,  p-value: < 2.2e-16
```

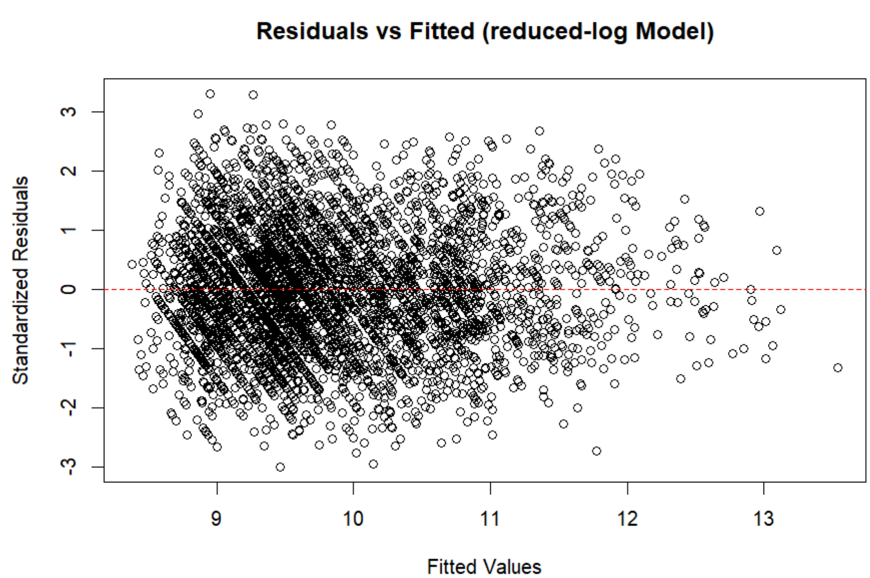
- **Residuals:** The residuals range between **-0.95 and 1.05**, which falls well within the acceptable range of -3 to 3. This confirms there are no extreme outliers affecting the rental rate predictions.
- **Significance of the Whole Model:** The model is highly significant with a p-value < **2.2e-16**, which is much smaller than the standard significance level ($\alpha = 0.05$). This means the regression model as a whole is statistically valid and the predictors have a real impact on rental rates.
- **Multiple R-squared = 0.8683:** This value indicates that 86.83% of the variation in the dependent variable (log-transformed rent) is explained by the independent variables included in the model. This is an exceptionally strong fit, meaning the predictors collectively account for most of the changes in rental prices.

```
vif(reduced_log_model)
                      GVIF Df GVIF^(1/(2*Df))
Size              2.962068  1        1.721066
BHK               3.464678  1        1.861365
Bathroom          3.935754  1        1.983874
City              2.838389  5        1.109960
Area.Type         1.687757  1        1.299137
Furnishing.Status 1.164235  2        1.038748
no_of_floors      1.755933  1        1.325116
Tenant.Preferred  1.270428  2        1.061665
Point.of.Contact  2.119327  1        1.455791
```

All VIF values are below the critical threshold of 10, indicating no severe multicollinearity in the reduced log model. The removal of the floor variable slightly improved the multicollinearity structure.

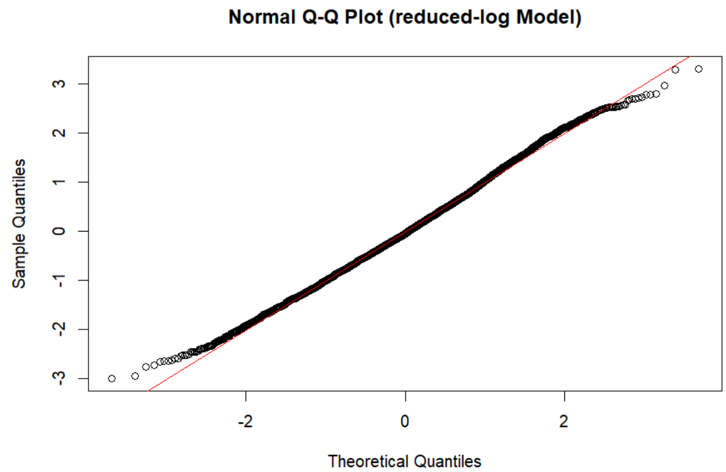
Check assumptions about error term in model1_reduced:

1- The variability in random error is constant (Homoscedasticity) and the Expected value of the error term equals zero, $\boldsymbol{E(\epsilon_i) = 0}$



The plot showed a random scatter of residuals around the horizontal axis with no clear pattern or funnel shape. This indicates that the variance of the residuals remains approximately constant across different levels of the fitted values. Hence, the assumption of **homoscedasticity** is considered to be satisfied for this model. This suggests that the model does not suffer from **heteroscedasticity**, supporting the validity of standard inference procedures**.**

2- check assumption normality of the error term for model1_reduced:



X-axis: quartiles of the normal distribution
Y-axis: quartiles of empirical data
Since the data take a linear form, this indicates that the data is normality.
This means that the assumptions of normality have not been violated.

```
shapiro.test(residuals(reduced_log_model))

        Shapiro-Wilk normality test

data:  residuals(reduced_log_model)
W = 0.99738, p-value = 6.044e-07
```

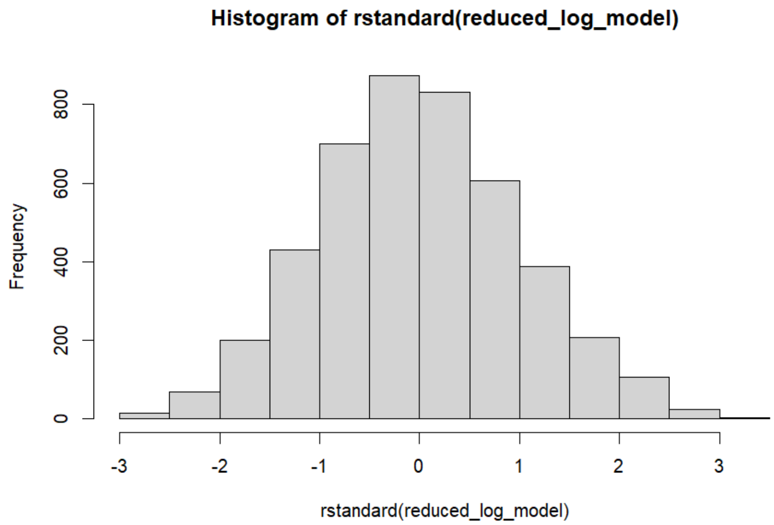
(H_0): Residuals follows a normal distribution.
(H_1): Residuals does not follow a normal distribution.

The test statistic **W = 0.99738** is very close to one, which typically suggests normality. However, the **p-value = 6.044e-07** is less than 0.05, so we reject the null hypothesis, indicating that normality is not met. This discrepancy arises because the **Shapiro-Wilk test** can produce misleading results with large sample sizes (greater than 50), where even small deviations from normality become statistically significant. However, the **Q-Q plot** clearly shows that the residuals follow a normal distribution, supporting the assumption of normality despite the p-value result.

```
ad.test(residuals(reduced_log_model))

        Anderson-Darling normality test

data:  residuals(reduced_log_model)
A = 2.6772, p-value = 9.567e-07
```

(H_0): Residuals follows a normal distribution.
(H_1): Residuals does not follow a normal distribution.

The test statistic **A = 2.6772**. Generally, a large value of this test statistic indicates non-normality. However, since the value here is relatively small, it suggests that the normality assumption is met. Despite this, the **p-value = 9.567e-07** is less than the significance level, which leads to rejecting the null hypothesis. This indicates the same issue observed with the Shapiro-Wilk test: with large sample sizes, the test may detect statistically significant deviations from normality even when they are not practically meaningful.



The graph appears **symmetric**, which is strong evidence that the **residuals follow a normal distribution**. Therefore, the **conflicting results** from the **Shapiro-Wilk** and **Anderson-Darling** normality tests can be reasonably **disregarded**.

3- check Are the errors independent (no auto -correlated between error) using, $\boldsymbol{cov}\ (\boldsymbol{\epsilon i}, \boldsymbol{\epsilon j}) = \mathbf{0}$:

```
durbinWatsonTest(reduced_log_model)

 lag Autocorrelation D-W Statistic p-value
   1     0.008726117      1.982468    0.53
 Alternative hypothesis: rho != 0
```

(H_0): There is NO first auto correlated between residuals $\rho 1=0$
(H_1): There is first auto correlated between residual $\rho 1 \neq 0$

The test statistic **D-W = 1.982468**, which is very close to 2. This indicates that there is no autocorrelation between the residuals. In addition, since the **p-value = 0.53** is greater than the significance level, we fail to reject the null hypothesis, further confirming that there is **no evidence of autocorrelation** among the residuals.

```
resettest(reduced_log_model, power = 2:3, type = "fitted")

        RESET test

data:  reduced_log_model
RESET = 4.1681, df1 = 2, df2 = 4434, p-value = 0.01554
```

(H_0): the model is correctly specified(no omitted variable bias)
(H_1): the model is misspecified.

Since the p-value is less than the 5% significance level, we reject the null hypothesis, indicating **moderate statistical evidence of model misspecification**. This suggests that the model might be missing important variables or that the functional form may not be entirely appropriate.

However, because the p-value is greater than 0.01, the evidence against the null hypothesis is not very strong at the 1% significance level.

The results of the reduced log model are found to be very close to those of the full log model with all floors included. Given this similarity and considering model parsimony, comparisons between models will be made using additional criteria such as the **Schwarz Criterion (BIC)**, **Akaike Information Criterion (AIC)**, and **ANOVA tests** to determine the best fitting and most efficient model.

log_reduced vs full_log model:

```
anova(reduced_log_model,log_model)
Analysis of Variance Table

Model 1: log(Rent) ~ Size + BHK + Bathroom + City + Area.Type + Furnishing.Status +
    no_of_floors + Tenant.Preferred + Point.of.Contact
Model 2: log(Rent) ~ Size + BHK + Bathroom + City + Area.Type + Furnishing.Status +
    floor + no_of_floors + Tenant.Preferred + Point.of.Contact
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1   4436 444.20
2   4435 444.01  1   0.18896 1.8875 0.1696
```

Using F-test:

(H_0): The parameter for floor variable= 0
(H_1): The parameter for floor variable ≠0

Since the F-statistic = **1.8875** and the p-value = **0.1696**, which is greater than the significance level of 0.05, we fail to reject the null hypothesis. This means that adding the floor variable did **not** significantly improve the model's performance.

Therefore, the **reduced model**—which excludes the *floor* variable—is considered the **better choice**.

```
AIC(reduced_log_model, log_model)
                  df      AIC
reduced_log_model 17 2407.064
log_model         18 2407.170
> BIC(reduced_log_model, log_model)
                  df      BIC
reduced_log_model 17 2515.883
log_model         18 2522.390
```

- **Akaike Information Criterion (AIC):**
  The reduced log model has an AIC of **2407.064**, which is slightly lower than the full log model's AIC of **2407.170**.
- **Bayesian Information Criterion (BIC):**
  The reduced log model has a BIC of **2515.883**, which is also lower than the full log model's BIC of **2522.390**.

Since both AIC and BIC values are lower for the reduced log model, this indicates that the reduced model provides a better balance of goodness-of-fit and model simplicity compared to the full model**. Therefore, the reduced log model is preferred as it achieves similar explanatory power with fewer parameters.**

The reduced log model is identified as the best main effects model among the four models considered. It achieves a strong balance between explanatory power and model simplicity, outperforming the others based on fit statistics and information criteria.

Testing the Significance of Individual Predictors:

β_0 (Intercept)
$H_0: \beta_0 = 0$ vs $H_1: \beta_0 \neq 0$
P-value = < 2e-16
Decision: Reject H_0 at 5% significance level.
Interpretation: The intercept (8.971) is statistically significant, indicating a meaningful baseline log-rent when all predictors are zero.

β_1 (Size)
$H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$
P-value = < 2e-16
Decision: Reject H_0 at 5% significance level.
Interpretation: Size is statistically significant ($\hat{\beta}_1$ = 4.614e-04), suggesting larger properties tend to have higher rent.

β_2 (BHK)
$H_0: \beta_2 = 0$ vs $H_1: \beta_2 \neq 0$
P-value = < 2e-16
Decision: Reject H_0 at 5% significance level.
Interpretation: BHK is significant ($\hat{\beta}_2$ = 2.047e-01), indicating more bedrooms increase rent.

β_3 (Bathroom)
$H_0: \beta_3 = 0$ vs $H_1: \beta_3 \neq 0$
P-value = < 2e-16
Decision: Reject H_0 at 5% significance level.
Interpretation: Bathrooms are significant ($\hat{\beta}_3$ = 1.556e-01), more bathrooms associate with higher rent.

β_{41} (City = Chennai)
P-value = 0.0751
Decision: Fail to reject H_0 at 5% significance level.
Interpretation: Chennai does not significantly affect rent compared to the reference city.

β_{42} (City = Delhi)
P-value = < 2e-16
Decision: Reject H_0 at 5% significance level.
Interpretation: Delhi increases rent significantly ($\hat{\beta}$ = 1.849e-01).

β_{43} (City = Hyderabad)
P-value = < 2e-16
Decision: Reject H_0 at 5% significance level.
Interpretation: Hyderabad decreases rent significantly ($\hat{\beta}$ = -1.346e-01).

β_{44} (City = Kolkata)
P-value = < 2e-16
Decision: Reject H_0 at 5% significance level.
Interpretation: Kolkata strongly decreases rent ($\hat{\beta}$ = -2.812e-01).

β_{45} (City = Mumbai)
P-value = < 2e-16
Decision: Reject H_0 at 5% significance level.
Interpretation: Mumbai strongly increases rent ($\hat{\beta}$ = 9.592e-01).

β_5 (Area.Type = Super Area)
P-value = 9.67e-05
Decision: Reject H_0 at 5% significance level.
Interpretation: Super Area is associated with lower rent ($\hat{\beta}$ = -4.815e-02).

β_{61} (Furnishing.Status = Semi-Furnished)
P-value = < 2e-16
Decision: Reject H_0 at 5% significance level.
Interpretation: Semi-furnished units have significantly lower rent ($\hat{\beta}$ = -1.795e-01) than furnished.

β_{62} (Furnishing.Status = Unfurnished)
P-value = < 2e-16
Decision: Reject H_0 at 5% significance level.
Interpretation: Unfurnished units reduce rent significantly ($\hat{\beta}$ = -2.941e-01).

β_7 (no_of_floors)
P-value = < 2e-16
Decision: Reject H_0 at 5% significance level.
Interpretation: Number of floors positively influences rent ($\hat{\beta}$ = 6.030e-03).

β_{81} (Tenant.Preferred = Bachelors/Family)
P-value = 0.0332
Decision: Reject H_0 at 5% significance level.
Interpretation: Tenant preference for Bachelors/Family is associated with slightly lower rent ($\hat{\beta}$ = -2.954e-02).

β_{82} (Tenant.Preferred = Family)
P-value = 1.85e-05
Decision: Reject H_0 at 5% significance level.
Interpretation: Renting only to families significantly lowers rent ($\hat{\beta}$ = -8.471e-02).

β_9 (Point.of.Contact = Contact Owner)
P-value = < 2e-16
Decision: Reject H_0 at 5% significance level.
Interpretation: Contacting owner directly is associated with lower rent ($\hat{\beta}$ = -3.250e-01).

Interpretation of Model Coefficients:

- **Intercept (β_0 = 8.971):**
  The average log-rent when all predictor variables are zero, i.e., the apartment is fully furnished, located in Bangalore, measured in carpet area, tenant preferred is bachelor, and the point of contact is an agent.
- **Size (β_1 = 0.0004614):**
  The average change in log-rent for each additional square meter of size, holding all other variables constant.
- **BHK (β_2 = 0.2047):**
  The average change in log-rent for each additional bedroom or kitchen, holding other variables constant.
- **Bathroom (β_3 = 0.1556):**
  The average change in log-rent for each additional bathroom, holding other variables constant.
- **City Chennai (β_{41} = -0.02765):**
  The difference in average log-rent between an apartment located in Chennai and one located in Bangalore (reference city), holding other variables constant.
- **City Delhi (β_{42} = 0.1849):**
  The difference in average log-rent between an apartment located in Delhi and Bangalore, holding other variables constant.
- **City Hyderabad (β_{43} = -0.1346):**
  The difference in average log-rent between an apartment located in Hyderabad and Bangalore, holding other variables constant.
- **City Kolkata (β_{44} = -0.2812):**
  The difference in average log-rent between an apartment located in Kolkata and Bangalore, holding other variables constant.
- **City Mumbai (β_{45} = 0.9592):**
  The difference in average log-rent between an apartment located in Mumbai and Bangalore, holding other variables constant.
- **Area.Type Super Area (β_5 = -0.04815):**
  The difference in average log-rent between apartments classified as "Super Area" compared to "Carpet Area," holding other variables constant.
- **Furnishing.Status Semi-Furnished (β_{61} = -0.1795):**
  The difference in average log-rent between semi-furnished and fully furnished apartments, holding other variables constant.
- **Furnishing.Status Unfurnished (β_{62} = -0.2941):**
  The difference in average log-rent between unfurnished and fully furnished apartments, holding other variables constant.
- **Number of Floors (β_7 = 0.00603):**
  The average change in log-rent for each additional floor, holding other variables constant.
- **Tenant.Preferred Bachelors/Family (β_{81} = -0.02954):**
  The difference in average log-rent between apartments preferred by bachelors/families compared to bachelors only, holding other variables constant.
- **Tenant.Preferred Family (β_{82} = -0.08471):**
  The difference in average log-rent between apartments preferred by families only compared to bachelors only, holding other variables constant.
- **Point.of.Contact Contact Owner (β_9 = -0.325):**
  The difference in average log-rent between apartments where the point of contact is the owner versus an agent, holding other variables constant.

Interaction model:

To address the research questions:

2. **Does the size of the property affect the rent price differently across different cities?**
3. **Are there differential effects of area type (Carpet vs. Super Area) on rent prices across cities?**

an interaction model was fitted including interaction terms between **City** and **Size** (to capture variation in price per square meter by city) and between **City** and **Area.Type** (to explore whether the impact of area type on rent differs by city). This approach allows us to test if the relationship between size and rent, as well as the influence of area type, depend on the city location.

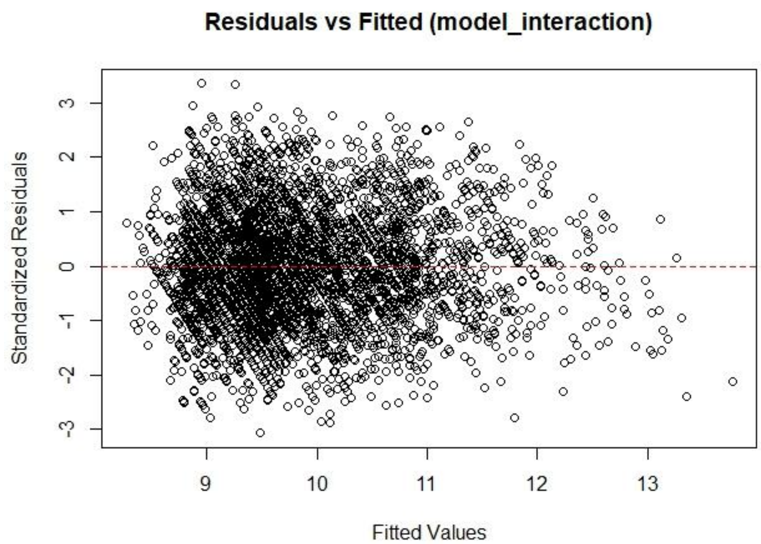
```
summary(model_interaction)

Call:
lm(formula = log(Rent) ~ Size + BHK + Bathroom + City + Area.Type +
    Furnishing.Status + no_of_floors + Tenant.Preferred + Point.of.Contact +
    Area.Type:City + Size:City, data = df_clean)

Residuals:
     Min       1Q   Median       3Q      Max
-0.95748 -0.21689 -0.00908  0.20252  1.06238

Coefficients:
                                    Estimate Std. Error t value Pr(>|t|)
(Intercept)                        9.015e+00  3.584e-02 251.559  < 2e-16 ***
Size                               4.881e-04  2.463e-05  19.819  < 2e-16 ***
BHK                                1.914e-01  1.137e-02  16.824  < 2e-16 ***
Bathroom                           1.528e-01  1.176e-02  12.995  < 2e-16 ***
CityChennai                       -2.358e-02  4.118e-02  -0.573 0.566980
CityDelhi                          1.203e-01  3.988e-02   3.017 0.002566 **
CityHyderabad                      5.054e-03  4.327e-02   0.117 0.907024
CityKolkata                       -3.701e-01  4.397e-02  -8.417  < 2e-16 ***
CityMumbai                         8.529e-01  3.788e-02  22.517  < 2e-16 ***
Area.TypeSuper Area               -8.426e-02  2.349e-02  -3.587 0.000338 ***
Furnishing.StatusSemi-Furnished   -1.769e-01  1.484e-02 -11.922  < 2e-16 ***
Furnishing.StatusUnfurnished      -2.969e-01  1.535e-02 -19.341  < 2e-16 ***
no_of_floors                       5.593e-03  7.299e-04   7.664 2.21e-14 ***
Tenant.PreferredBachelors/Family  -3.160e-02  1.386e-02  -2.280 0.022679 *
Tenant.PreferredFamily            -7.965e-02  1.977e-02  -4.029 5.69e-05 ***
Point.of.ContactContact Owner     -3.423e-01  1.554e-02 -22.036  < 2e-16 ***
CityChennai:Area.TypeSuper Area    3.204e-02  3.209e-02   0.998 0.318143
CityDelhi:Area.TypeSuper Area      1.254e-01  3.578e-02   3.504 0.000463 ***
CityHyderabad:Area.TypeSuper Area  8.347e-03  3.371e-02   0.248 0.804449
CityKolkata:Area.TypeSuper Area    7.464e-03  3.655e-02   0.204 0.838190
CityMumbai:Area.TypeSuper Area     9.323e-02  3.866e-02   2.412 0.015918 *
Size:CityChennai                  -2.156e-05  2.992e-05  -0.721 0.471190
Size:CityDelhi                     2.764e-06  3.059e-05   0.090 0.928017
Size:CityHyderabad                -1.240e-04  2.817e-05  -4.402 1.10e-05 ***
Size:CityKolkata                   1.123e-04  4.321e-05   2.598 0.009407 **
Size:CityMumbai                    8.649e-05  3.063e-05   2.824 0.004764 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.314 on 4426 degrees of freedom
Multiple R-squared:  0.8711,	Adjusted R-squared:  0.8703
```

```
F-statistic:  1196 on 25 and 4426 DF,  p-value: < 2.2e-16
```

- **Overall Fit**:
  - Adj. R-squared = 0.8703 → The model explains 87% of variance in log(Rent)
  - F-statistic = 1196 ($p < 2.2e-16$) → Highly significant overall model
- **Residuals**:
  - Range: -0.957 to 1.06 → Well within normal range (-3 to 3)
  - Median near zero (-0.00908) → No systematic bias

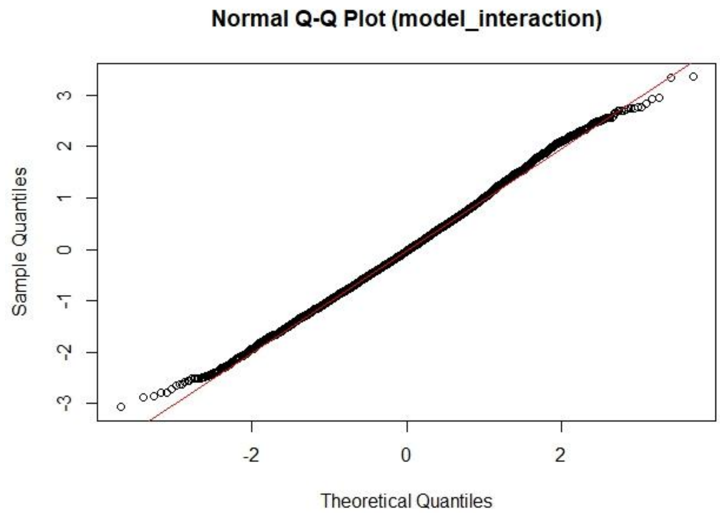
Check assumptions about error term in interaction term:

1- The variability in random error is constant (Homoscedasticity) and the Expected value of the error term equals zero, $\boldsymbol{E(\epsilon_i) = 0}$



The plot showed a random scatter of residuals around the horizontal axis with no clear pattern or. This indicates that the variance of the residuals remains approximately constant across different levels of the fitted values. Hence, the assumption of **homoscedasticity** is considered to be satisfied for this model. This suggests that the model does not suffer from **heteroscedasticity**, supporting the validity of standard inference procedures**.**

2- check assumption normality of the error term for initial model:



X-axis: quartiles of the normal distribution
Y-axis: quartiles of empirical data
Since the data take a linear form, this indicates that the data is normality. This means that the assumptions of normality have not been violated.

```
> shapiro.test(residuals(model_interaction))

        Shapiro-Wilk normality test

data:  residuals(model_interaction)
W = 0.99809, p-value = 2.83e-05
```

(H_0): The residuals follow a normal distribution.
(H_1): The residuals do not follow a normal distribution.

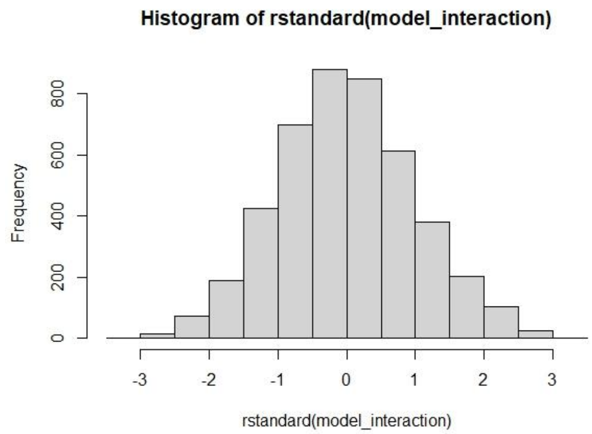
The test statistic W = **0.99809**. Since this value is very close to 1, it suggests evidence of normality. However, the p-value = **2.83e-05** is less than the significance level of 0.05, so we reject the null hypothesis, indicating that the assumption of normality is not met.

This discrepancy between the test statistic and the p-value arises because the Shapiro-Wilk test may not provide clear results when the sample size exceeds 50. Nonetheless, the Q-Q plot visually indicated that the residuals approximately follow a normal distribution.

```
ad.test(residuals(model_interaction))

        Anderson-Darling normality test

data:  residuals(model_interaction)
A = 1.9552, p-value = 5.55e-05
```

(H_0): Residuals follows a normal distribution.
(H_1): Residuals does not follow a normal distribution.

The test statistic A = 1.9552, Since the large value of the selection statistic indicates non-normality, and here the value of the test statistics is very small, this indicates that the normality assumption is met. Since the p_value is less than the significant level, we find that there is the same problem exists in the Shapiro-Wilk test.



The graph is symmetric so this is evidence that the residuals follow a normal distribution so we can ignore the conflicting values for Shapiro-Wilk test and Anderson-Darling normal.

3- check Are the errors independent (no auto -correlated between error) using, $\boldsymbol{cov}\,(\boldsymbol{\epsilon i}, \boldsymbol{\epsilon j}) = \mathbf{0}$:

```
durbinWatsonTest(model_interaction)
 lag Autocorrelation D-W Statistic p-value
   1        0.010486      1.978968   0.406
 Alternative hypothesis: rho != 0
```

(H_0): There is NO first auto correlated between residuals $\rho1=0$

(H_1): There is first auto correlated between residual $\rho1 \neq 0$

The test statistic D-W = 1.978968, it is clear that the value of the test statistic is very close to 2, which is evidence that there is no autocorrelation between residuals. Also, since the value of the p-value is greater than the level of significance, the null hypothesis will not be rejected, which confirms that there is no autocorrelation between residuals.

```
resettest(model_interaction, power = 2:3, type = "fitted")

        RESET test

data:  model_interaction
RESET = 12.839, df1 = 2, df2 = 4424, p-value = 2.757e-06
```

(H_0): the model is correctly specified (no omitted variable bias)

(H_1): the model is misspecified.

Since p-value = 2.757e-06 is smaller than the significance level of 0.05, then reject the null hypothesis. Although the null hypothesis was rejected due to the increase in categorical variables.

```
anova(reduced_log_model,model_interaction)
Analysis of Variance Table

Model 1: log(Rent) ~ Size + BHK + Bathroom + City + Area.Type + Furnishing.Status + 
    no_of_floors + Tenant.Preferred + Point.of.Contact
Model 2: log(Rent) ~ Size + BHK + Bathroom + City + Area.Type + Furnishing.Status + 
    no_of_floors + Tenant.Preferred + Point.of.Contact + Area.Type:City + 
    Size:City
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
```

```
1   4436 444.20
2   4426 436.41 10     7.787 7.8975 1.057e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Using f-test:

(H_0): all interaction petameters = 0
(H_1): at least one is different

Since the value F = 7.8975 is greater than the test statistic F and the p-value =1.057e-12 is less than the significance level of 0.05, the null hypothesis will be rejected, which means that adding the interaction variables greatly improved the performance of the model.

Testing the Significance of Individual Predictors:

H0: $\boldsymbol{\beta 0} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 0} \neq \mathbf{0}$
P-value = < 2e-16 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: The intercept is statistically significant, indicating a meaningful baseline log rent price when all predictors are zero.

H0: $\boldsymbol{\beta 1} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 1} \neq \mathbf{0}$
P-value = < 2e-16 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: Size is statistically significant (β_1 = 0.0004881), implying that larger properties tend to have higher log rent prices.

H0: $\boldsymbol{\beta 2} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 2} \neq \mathbf{0}$
P-value = < 2e-16 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: BHK is a significant predictor (β_2 = 0.1914), suggesting more bedrooms are associated with higher rent.

H0: $\boldsymbol{\beta 3} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 3} \neq \mathbf{0}$
P-value = < 2e-16 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: Bathroom count is statistically significant (β_3 = 0.1528), indicating that properties with more bathrooms tend to be priced higher.

H0: $\boldsymbol{\beta 4} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 4} \neq \mathbf{0}$
P-value = 0.566980 > 0.05
Decision: The null hypothesis will not be rejected at significance level = 5%.
Interpretation: CityChennai is not statistically significant (β_4 = -0.02358), indicating no meaningful difference in log rent compared to the base city.

H0: $\boldsymbol{\beta 5} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 5} \neq \mathbf{0}$
P-value = 0.002566 < 0.05

Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: CityDelhi is statistically significant ($\hat{\beta}_5 = 0.1203$), suggesting rents in Delhi are significantly higher than the base city.

$\mathbf{H0}$: $\boldsymbol{\beta 6} = \mathbf{0}$ vs $\mathbf{H1}$: $\boldsymbol{\beta 6} \neq \mathbf{0}$
P-value = 0.907024 > 0.05
Decision: The null hypothesis will not be rejected at significance level = 5%.
Interpretation: CityHyderabad is not statistically significant ($\hat{\beta}_6 = 0.005054$), indicating rents are not meaningfully different from the base city.

$\mathbf{H0}$: $\boldsymbol{\beta 7} = \mathbf{0}$ vs $\mathbf{H1}$: $\boldsymbol{\beta 7} \neq \mathbf{0}$
P-value = < 2e-16 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: CityKolkata is statistically significant ($\hat{\beta}_7 = -0.3701$), indicating that rents are significantly lower in Kolkata compared to the base city.

$\mathbf{H0}$: $\boldsymbol{\beta 8} = \mathbf{0}$ vs $\mathbf{H1}$: $\boldsymbol{\beta 8} \neq \mathbf{0}$
P-value = < 2e-16 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: CityMumbai is statistically significant ($\hat{\beta}_8 = 0.8529$), indicating higher rents in Mumbai.

$\mathbf{H0}$: $\boldsymbol{\beta 9} = \mathbf{0}$ vs $\mathbf{H1}$: $\boldsymbol{\beta 9} \neq \mathbf{0}$
P-value = 0.000338 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: Area Type "Super Area" is statistically significant ($\hat{\beta}_9 = -0.08426$), suggesting that this area type is associated with lower rent.

$\mathbf{H0}$: $\boldsymbol{\beta 10} = \mathbf{0}$ vs $\mathbf{H1}$: $\boldsymbol{\beta 10} \neq \mathbf{0}$
P-value = < 2e-16 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: Semi-Furnished status is statistically significant ($\hat{\beta}_{10} = -0.1769$), implying lower rents than Fully-Furnished.

$\mathbf{H0}$: $\boldsymbol{\beta 11} = \mathbf{0}$ vs $\mathbf{H1}$: $\boldsymbol{\beta 11} \neq \mathbf{0}$
P-value = < 2e-16 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: Unfurnished properties also show significantly lower rent ($\hat{\beta}_{11} = -0.2969$).

$\mathbf{H0}$: $\boldsymbol{\beta 12} = \mathbf{0}$ vs $\mathbf{H1}$: $\boldsymbol{\beta 12} \neq \mathbf{0}$
P-value = 2.21e-14 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: Number of floors is statistically significant ($\hat{\beta}_{12} = 0.005593$), indicating that higher floors may correspond with higher rent.

$\mathbf{H0}$: $\boldsymbol{\beta 13} = \mathbf{0}$ vs $\mathbf{H1}$: $\boldsymbol{\beta 13} \neq \mathbf{0}$
P-value = 0.022679 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: Tenant preference "Bachelors/Family" is statistically significant ($\hat{\beta}_{13} = -0.0316$), indicating this preference is associated with lower rent.

H0: $\boldsymbol{\beta 14} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 14} \neq \mathbf{0}$
P-value = 5.69e-05 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: Tenant preference "Family" is statistically significant ($\hat{\beta}_{14}$ = -0.07965), indicating a negative effect on rent.

H0: $\boldsymbol{\beta 15} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 15} \neq \mathbf{0}$
P-value = < 2e-16 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: Contacting the owner directly (Point of Contact = Owner) is significant ($\hat{\beta}_{15}$ = -0.3423), suggesting it is associated with lower rent.

H0: $\boldsymbol{\beta 16} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 16} \neq \mathbf{0}$
P-value = 0.318143 > 0.05
Decision: The null hypothesis will not be rejected at significance level = 5%.
Interpretation: The interaction between CityChennai and Area Type "Super Area" is not statistically significant ($\hat{\beta}_{16}$ = 0.03204).

H0: $\boldsymbol{\beta 17} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 17} \neq \mathbf{0}$
P-value = 0.000463 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: The interaction between CityDelhi and "Super Area" is statistically significant ($\hat{\beta}_{17}$ = 0.1254), indicating higher rent in this combination.

H0: $\boldsymbol{\beta 18} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 18} \neq \mathbf{0}$
P-value = 0.804449 > 0.05
Decision: The null hypothesis will not be rejected at significance level = 5%.
Interpretation: The interaction between CityHyderabad and "Super Area" is not significant ($\hat{\beta}_{18}$ = 0.008347).

H0: $\boldsymbol{\beta 19} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 19} \neq \mathbf{0}$
P-value = 0.838190 > 0.05
Decision: The null hypothesis will not be rejected at significance level = 5%.
Interpretation: The interaction between CityKolkata and "Super Area" is not significant ($\hat{\beta}_{19}$ = 0.007464).

H0: $\boldsymbol{\beta 20} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 20} \neq \mathbf{0}$
P-value = 0.015918 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: The interaction between CityMumbai and "Super Area" is statistically significant ($\hat{\beta}_{20}$ = 0.09323), suggesting higher rents.

H0: $\boldsymbol{\beta 21} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 21} \neq \mathbf{0}$
P-value = 0.471190 > 0.05
Decision: The null hypothesis will not be rejected at significance level = 5%.
Interpretation: The interaction between Size and CityChennai is not significant ($\hat{\beta}_{21}$ = -0.00002156).

H0: $\boldsymbol{\beta 22} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 22} \neq \mathbf{0}$
P-value = 0.928017 > 0.05
Decision: The null hypothesis will not be rejected at significance level = 5%.
Interpretation: The interaction between Size and CityDelhi is not significant ($\hat{\beta}_{22}$ = 0.000002764).

H0: $\boldsymbol{\beta 23} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 23} \neq \mathbf{0}$
P-value = 1.10e-05 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: The interaction between Size and CityHyderabad is significant (β_{23} = -0.000124), indicating a decreasing effect.

H0: $\boldsymbol{\beta 24} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 24} \neq \mathbf{0}$
P-value = 0.009407 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: The interaction between Size and CityKolkata is significant (β_{24} = 0.0001123), suggesting higher rents.

H0: $\boldsymbol{\beta 25} = \mathbf{0}$ vs **H1**: $\boldsymbol{\beta 25} \neq \mathbf{0}$
P-value = 0.004764 < 0.05
Decision: The null hypothesis will be rejected at significance level = 5%.
Interpretation: The interaction between Size and CityMumbai is statistically significant (β_{25} = 0.00008649), also suggesting higher rents.

Interpretation of interaction coefficients:

1. **CityDelhi:Area.TypeSuper Area = 0.1254**
   The difference in average log(rent) between Delhi and the base city (Bangalore), when the area type is Super Area (compared to Carpet Area), is **0.1254**.
2. **CityMumbai:Area.TypeSuper Area = 0.09323**
   The difference in average log(rent) between Mumbai and Bangalore, when the area type is Super Area (compared to Carpet Area), is **0.09323**.
3. **Size:CityHyderabad = -0.000124**
   Difference in average log(rent) between Hyderabad and Bangalore when size increases by 1 square meter is **-0.000124**.
4. **Size:CityKolkata = 0.0001123**
   Difference in average log(rent) between Kolkata and Bangalore when size increases by 1 square meter is **0.0001123**.
5. **Size:CityMumbai = 0.00008649**
   Difference in average log(rent) between Mumbai and Bangalore when size increases by 1 square meter is **0.00008649**.

Conclusion:

This econometrics project aimed to analyze the determinants of house rental prices in major Indian cities using a dataset from Kaggle. The study addressed three primary research questions:

1. What are the key factors influencing house rent prices, and which factors have the most significant impact?

Variable	Coefficient	% Change in Rent	Interpretation
CityMumbai	0.9592	+161.15%	Rent in Mumbai is ~161.15% higher than the reference city
BHK	0.2047	+22.67%	Each additional bedroom increases rent by ~22.67%

Variable	Coefficient	% Change in Rent	Interpretation
CityDelhi	0.1849	+20.29%	Rent in Delhi is ~20.29% higher than the reference city
Bathroom	0.1556	+16.88%	Each additional bathroom increases rent by ~16.88%
no_of_floors	0.00603	+0.60%	Each additional floor in the building increases rent by ~0.60%
Size	0.0004614	+0.05%	Each additional sq ft increases rent by ~0.05%
CityChennai	-0.02765	−2.73%	Rent in Chennai is ~2.73% lower than the reference city
Tenant.PreferredBachelors/Family	-0.02954	−2.91%	Bachelors/Family preference lowers rent by ~2.91% compared to Family preference
Area.TypeSuper Area	-0.04815	−4.70%	Super Area type is associated with ~4.70% lower rent than Carpet Area
Tenant.PreferredFamily	-0.08471	−8.13%	Family-only preference lowers rent by ~8.13% compared to Bachelors
CityHyderabad	-0.1346	−12.58%	Rent in Hyderabad is ~12.58% lower than the reference city
Furnishing.StatusSemi-Furnished	-0.1795	−16.41%	Semi-furnished properties rent for ~16.41% less than furnished
CityKolkata	-0.2812	−24.43%	Rent in Kolkata is ~24.43% lower than the reference city
Furnishing.StatusUnfurnished	-0.2941	−25.43%	Unfurnished properties rent for ~25.43% less than furnished
Point.of.ContactContact Owner	-0.3250	−27.76%	Properties where the contact is the owner rent for ~27.76% less than through an agent

All interpretations with holding other variables constant, Percentage change=(exp(β)−1) ×100

The city where the property is located is the most significant factor influencing house rent prices. For example, Mumbai increases rent by approximately **161%** compared to the reference city, while Kolkata and Hyderabad reduce rent by about **24.4%** and **12.6%**, respectively. Delhi also raises rent by around **20.3%**.

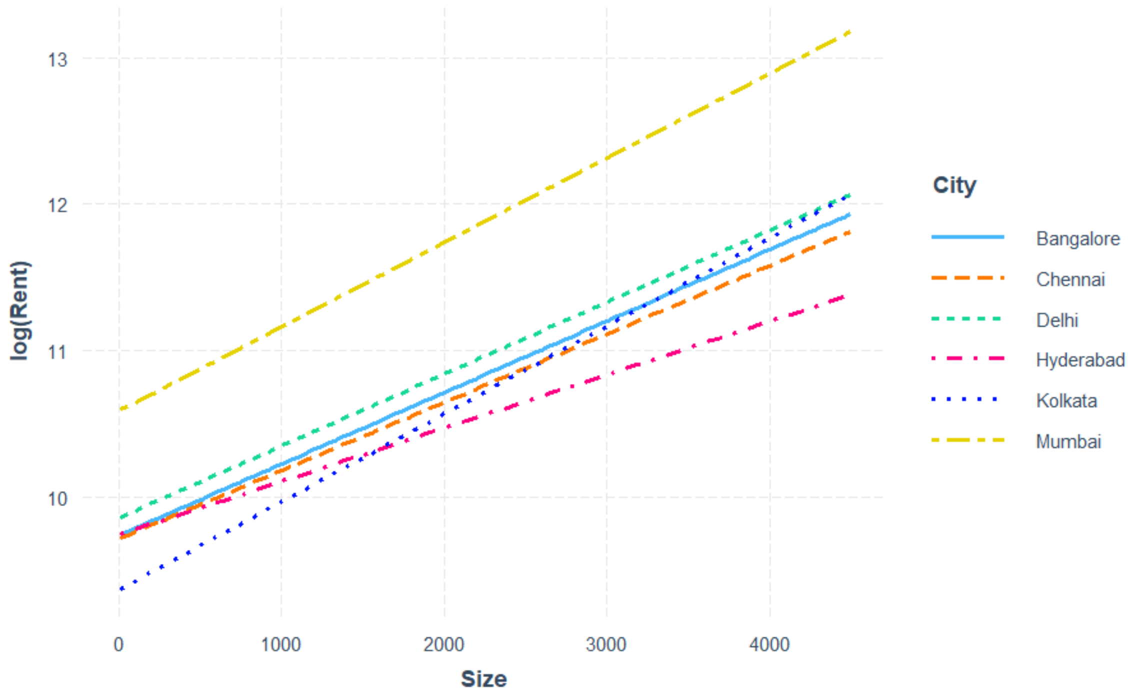
This shows that location plays a major role in determining rental values. Property features also have a strong impact on rent prices. The number of bedrooms (BHK) is particularly important, with each additional bedroom increasing rent by about **22.7%**. Bathrooms also matter, as each additional bathroom increases rent by roughly **16.9%**. Property size has a positive but smaller effect, with each unit increase in size raising rent by approximately **0.05%**. Furnishing status significantly affects rent as well. Unfurnished properties reduce rent by about **25.4%**, while semi-furnished properties reduce rent by approximately **16.4%**. This indicates that fully furnished properties generally command higher rents in the market.

Other factors like the number of floors in the building contribute positively to rent prices, with each additional floor increasing rent by around **0.6%**. Interestingly, when the point of contact for the rental is the owner, rent decreases by approximately **27.8%**, possibly due to better negotiation or absence of intermediaries. Tenant preferences also influence rent, where properties preferred for families reduce rent by around **8.1%**, and those preferred for bachelors/families combined reduce rent by about **2.9%**. Lastly, the type of area labeled as "Super Area" has a small negative effect on rent, decreasing it by around **4.7%**.

In summary, the strongest factors affecting rent prices are city location, furnishing status,the number of bedrooms and bathrooms and point of contact. Big cities like Mumbai and Delhi command substantially higher rents compared to other cities, and other factors like direct owner contact and tenant preferences also noticeably influence rental prices.

2. Does the size of the property affect the rent price differently across different cities?

Variable	Coefficient	% Change in Rent	Interpretation
Size:CityHyderabad	-0.000124	-0.0124%	For Hyderabad, each unit increase in Size decreases rent by ~0.0124% compared to the reference city.
Size:CityKolkata	0.0001123	+0.0112%	For Kolkata, each unit increase in Size increases rent by ~0.0112% more than the reference city.
Size:CityMumbai	0.00008649	+0.00865%	For Mumbai, each unit increase in Size increases rent by ~0.00865% more than the reference city.



The coefficient for **Size:CityHyderabad** is negative (-0.000124), suggesting that in Hyderabad, each additional square foot is associated with a very slight decrease in rent of about **0.0124%** compared to the baseline city. In contrast, **Size:CityKolkata** and **Size:CityMumbai** have positive coefficients (0.0001123 and 0.00008649 respectively), indicating slightly stronger positive effects of size on rent in these cities—roughly **+0.0112%** and **+0.00865%** per additional square foot, compared to the baseline. There are differences in the intercepts across cities, which were considered in the main effects model, where Delhi has the highest intercept and Kolkata the lowest compared to Bangalore as illustrated in the table.

However, it's important to note that these interaction effects between size and city are quite small in magnitude, and for cities other than Hyderabad, Kolkata, and Mumbai, the size interaction terms were found to be statistically insignificant. This means that for most cities, the relationship between size and rent remains similar to the baseline, implying a mostly uniform effect of size across locations.

In practical terms, while size generally increases rent, the rate of increase is fairly consistent across different cities, with only minor variations that may not be statistically robust. This uniformity simplifies modeling and interpretation, suggesting that size is a fundamental driver of rent across all regions, with only slight local modifications.

3. Are there differential effects of area type (Carpet vs. Super Area) on rent prices across cities?

Variable	Coefficient	% Change in Rent	Interpretation
CityDelhi:Area.TypeSuper Area	0.1254	+13.3%	In Delhi, being Super Area increases rent by ~13.3% compared to the reference city with carpet area.
CityMumbai:Area.TypeSuper Area	0.09323	+9.76%	In Mumbai, being Super Area increases rent by ~9.76% compared to the reference city with carpet area.

data suggests that in major metropolitan cities like Delhi and Mumbai, the Super Area classification plays a more prominent role in increasing rental prices. This premium likely reflects the inclusion of additional spaces or amenities - such as balconies, gardens, and common areas- that are counted within the Super Area measurement. Tenants in these high-demand markets are willing to pay more for these extra features. For urban planners, investors, and renters, understanding this distinction helps explain why rents might be higher for Super Area properties in certain cities and underscores the importance of standardized area measurements in rent pricing.

For real estate developers in India, it's crucial to design properties that reflect local market preferences. Prioritize adding value through Super Area features like balconies and gardens in high-demand cities such as Mumbai and Delhi, where tenants are willing to pay a premium. Focus on optimizing the number of bedrooms and bathrooms and offer fully furnished options to maximize rent. Always consider city-specific dynamics, as size and area type impacts differ across regions. By aligning your developments with these insights, you can better meet tenant expectations and achieve higher rental yields.

APPENDIX:

Link of the data: [House Rent Prediction Dataset]

```r
# Load necessary libraries
library(dplyr)
library(ggplot2)
library(stringr)
library(car)
library(lmtest)
library(interactions)
library(nortest)

# Read the data
data <- read.csv("D:/ECOMETRICS PROJECT/House_Rent_Dataset.csv")
head(data)
str(data)

# Convert specified columns to factors
factor_cols <- c("Area.Type", "City", "Furnishing.Status",
                 "Tenant.Preferred", "Point.of.Contact")
data[factor_cols] <- lapply(data[factor_cols], as.factor)

df <- data.frame(data)
# Split Floor column and select - FIXED VERSION
df <- df %>%
  mutate(
    no_of_floors = as.numeric(str_extract(Floor, "(?<=out of )\\d+")),
    floor = case_when(
      grepl("Ground", Floor) ~ 0,
      grepl("Upper Basement", Floor, ignore.case = TRUE) ~ -1,
      grepl("Lower Basement", Floor, ignore.case = TRUE) ~ -2,
      TRUE ~ as.numeric(str_extract(Floor, "^\\d+"))
    )
  ) %>%
  dplyr::select(BHK, Rent, Size, no_of_floors, floor, everything(), -Floor)
#remove sparse categories
head(df)
summary(df)
df <- df %>%
  filter(
    Area.Type != "Built Area",
    Point.of.Contact != "Contact Builder",
    !is.na(no_of_floors),          # Remove NA in floor counts
    Bathroom < 6                   # Remove rows with 6 or more bathrooms
  ) %>%
  droplevels()

#remove outliers:
model <- lm(log(Rent) ~ Size + BHK + no_of_floors + floor + Area.Type +
              City + Furnishing.Status
```

```r
            + Tenant.Preferred + Bathroom + Point.of.Contact
            ,data = df)
n <- nrow(df)
df_clean <- df %>%
  mutate(cooks_d = cooks.distance(model)) %>%
  filter(cooks_d <= 4/n) %>%
  dplyr::select(-cooks_d)
summary(df_clean)

#-----------------------------------------------------------------------------------
# Scatterplots for quantitative variables
plot(df_clean$Size, df_clean$Rent,
     main="Rent vs Size (Cleaned Data)",
     xlab="Size", ylab="Rent",
     pch=20, col="gray50")

plot(df_clean$BHK, df_clean$Rent,
     main="Rent vs BHK (Cleaned Data)",
     xlab="BHK", ylab="Rent)",
     pch=20, col="gray50")

plot(df_clean$Bathroom, df_clean$Rent,
     main="Rent vs Bathroom (Cleaned Data)",
     xlab="Bathroom", ylab="Rent",
     pch=20, col="gray50")

plot(df_clean$no_of_floors, df_clean$Rent,
     main="Rent vs no_of_floors (Cleaned Data)",
     xlab="no_of_floors", ylab="Rent",
     pch=20, col="gray50")

plot(df_clean$floor, df_clean$Rent,
     main="Rent vs floor (Cleaned Data)",
     xlab="floor", ylab="Rent",
     pch=20, col="gray50")

# Boxplots for categorical variables
boxplot(log(Rent) ~ City, data=df_clean,
        main="Rent by City (Cleaned Data)",
        xlab="City", ylab="log(Rent)",
        col="lightblue")

boxplot(log(Rent) ~ Area.Type, data=df_clean,
        main="Rent by Area Type (Cleaned Data)",
        xlab="Area Type", ylab="log(Rent)",
        col="lightblue")

boxplot(log(Rent) ~ Furnishing.Status, data=df_clean,
        main="Rent by Furnishing (Cleaned Data)",
        xlab="Furnishing Status", ylab="log(Rent)",
```

```
        col="lightblue")

boxplot(log(Rent) ~ Tenant.Preferred, data=df_clean,
        main="Rent by Tenant Preference (Cleaned Data)",
        xlab="Tenant Preferred", ylab="log(Rent)",
        col="lightblue")

boxplot(log(Rent) ~ Point.of.Contact, data=df_clean,
        main="Rent by Contact Type (Cleaned Data)",
        xlab="Point of Contact", ylab="Rent",
        col="lightblue")
#------------------------------------------------------------------------------------
# Initial regression model
model <- lm(Rent ~     Size + BHK + Bathroom + City + Area.Type +
                      Furnishing.Status + floor+ no_of_floors + Tenant.Preferred +
                      Point.of.Contact , data = df_clean)
# Display model summary
summary(model)
#test mullticollinerity
vif(model)
# Residuals vs Fitted
plot(rstandard(model) ~ fitted(model), main = "Residuals vs Fitted (Initial Model)",
     xlab = "Fitted Values", ylab = "Standardized Residuals")
abline(h = 0, lty = 2, col = "red")
# Q-Q plot
qqnorm(rstandard(model), main = "Normal Q-Q Plot (Initial Model)")
qqline(rstandard(model), col = "red")
#auto colleration of residuals test
durbinWatsonTest(model)
#Model misspecification test
resettest(model, power = 2:3, type = "fitted")
#normality tests
shapiro.test(residuals(model))
ad.test(residuals(model))
#histogram residuals
hist(rstandard(model))
#-------------------------------------------------------------------------------------
# Log regression model
log_model <- lm(log(Rent) ~ Size + BHK + Bathroom + City + Area.Type +
              Furnishing.Status + floor+ no_of_floors + Tenant.Preferred +
              Point.of.Contact , data = df_clean)
# Display model summary
summary(log_model)
#test mullticollinerity
vif(log_model)
# Residuals vs Fitted
plot(rstandard(log_model) ~ fitted(log_model), main = "Residuals vs Fitted (log Model)",
     xlab = "Fitted Values", ylab = "Standardized Residuals")
abline(h = 0, lty = 2, col = "red")
# Q-Q plot
```

```r
qqnorm(rstandard(log_model), main = "Normal Q-Q Plot (log Model)")
qqline(rstandard(log_model), col = "red")
#auto colleration of residuals test
durbinWatsonTest(log_model)
#Model misspecification test
resettest(log_model, power = 2:3, type = "fitted")
#normality tests
shapiro.test(residuals(log_model))
ad.test(residuals(log_model))
#histogram residuals
hist(rstandard(log_model))

#-------------------------------------------------------------------------------------------
#log_log_model
log_log_model <- lm(log(Rent) ~ log(Size) + BHK + Bathroom + City + Area.Type +
                  Furnishing.Status +floor + no_of_floors + Tenant.Preferred +
                  Point.of.Contact , data = df_clean)
vif(log_log_model)
summary(log_log_model)
# Residuals vs Fitted
plot(rstandard(log_log_model) ~ fitted(log_log_model), main = "Residuals vs Fitted (log-log
Model)",
     xlab = "Fitted Values", ylab = "Standardized Residuals")
abline(h = 0, lty = 2, col = "red")
# Q-Q plot
qqnorm(rstandard(log_log_model), main = "Normal Q-Q Plot (log-log Model)")
qqline(rstandard(log_log_model), col = "red")
#auto colleration of residuals test
durbinWatsonTest(log_log_model)
#Model misspecification test
resettest(log_log_model, power = 2:3, type = "fitted")
#normality tests
shapiro.test(residuals(log_log_model))
ad.test(residuals(log_log_model))
#histogram residuals
hist(rstandard(log_log_model))
#compare between two models
AIC(log_log_model, log_model)
BIC(log_log_model, log_model)
#-------------------------------------------------------------------------------------
reduced_log_model <- lm(log(Rent) ~ Size + BHK + Bathroom + City + Area.Type +
                      Furnishing.Status + no_of_floors + Tenant.Preferred +
                      Point.of.Contact , data = df_clean)
vif(reduced_log_model)
summary(reduced_log_model)
# Residuals vs Fitted
plot(rstandard(reduced_log_model) ~ fitted(reduced_log_model), main = "Residuals vs Fitted
(reduced-log Model)",
     xlab = "Fitted Values", ylab = "Standardized Residuals")
abline(h = 0, lty = 2, col = "red")
```

```
# Q-Q plot
qqnorm(rstandard(reduced_log_model), main = "Normal Q-Q Plot (reduced-log Model)")
qqline(rstandard(reduced_log_model), col = "red")
#auto colleration of residuals test
durbinWatsonTest(reduced_log_model)
#Model misspecification test
resettest(reduced_log_model, power = 2:3, type = "fitted")
#normality tests
shapiro.test(residuals(reduced_log_model))
ad.test(residuals(reduced_log_model))
#histogram residuals
hist(rstandard(reduced_log_model))
#compare between two models
anova(reduced_log_model,log_model)
#compare between two models
AIC(reduced_log_model, log_model)
BIC(reduced_log_model, log_model)
#-------------------------------------------------------------------------------------
#interaction_model
model_interaction <- lm(
  log(Rent) ~ Size + BHK + Bathroom + City + Area.Type +
    Furnishing.Status + no_of_floors + Tenant.Preferred + Point.of.Contact +
    Area.Type:City + Size:City,
  data = df_clean
)
summary(model_interaction)

# Residuals vs Fitted
plot(rstandard(model_interaction) ~ fitted(model_interaction), main = "Residuals vs Fitted
(interaction Model)",
     xlab = "Fitted Values", ylab = "Standardized Residuals")
abline(h = 0, lty = 2, col = "red")
# Q-Q plot
qqnorm(rstandard(model_interaction), main = "Normal Q-Q Plot (interaction Model)")
qqline(rstandard(model_interaction), col = "red")
#auto colleration of residuals test
durbinWatsonTest(model_interaction)
#Model misspecification test
resettest(model_interaction, power = 2:3, type = "fitted")
#normality tests
shapiro.test(residuals(model_interaction))
ad.test(residuals(model_interaction))
#histogram residuals
hist(rstandard(model_interaction))
#check if the interaction model better than the main effect model
anova(reduced_log_model,model_interaction)
interact_plot(model_interaction, pred= Size, modx= City)
```